THE LENGTH-OVER-DIAMETER RATIO OF SELF-APPROACHING CURVES

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Abstract. A rectifiable plane curve is self-approaching if its distance to every later point of itself is nonincreasing. Let C_{diam} be the supremum of length over diameter among such curves. The hull inequality of Manselli and Pucci gives $C_{\mathrm{diam}} \le \pi$. We prove

$$1 + \frac{2\pi}{3} \;\le\; C_{\mathrm{diam}} \;\le\; \frac{2\pi}{3} + 4\sin\frac{\pi}{12},$$

that is, $3.0943\ldots \le C_{\mathrm{diam}} \le 3.1296\ldots$. The upper bound rests on an exact formula for the perimeter of the convex hull of a self-approaching curve, whose defect vanishes exactly where the rest of the curve fills a right angle with the tangent along one side, and on an exact dual certificate for the largest perimeter of a convex body of diameter one with a right-angled corner. The trimmed Reuleaux triangle attains it, so these properties of the hull give no more; but it is not the hull of a self-approaching curve, and the upper bound is strict. The lower bound is the exact supremum over curves that begin on the circle about their last diametral partner. To prove it we construct self-approaching curves in a circular segment with corners of angle $\pi/6$ whose length tends to the perimeter of the segment. They track its outer arc, which no self-approaching curve can follow without slack, by oscillating along inscribed chords. The oscillations are cycloidal cascades whose tops are cut off by chords. We show that they attain the largest possible multiplicity $1 + \cos\Theta$ of an oscillation whose later points lie in a wedge of angle Θ on one side, and that, unlike the classical cascade, they can be followed by a turn.

1. Introduction

A curve $C\colon [0,1] \to \mathbb{R}^2$ is *self-approaching* if

$$|C(t_2) - C(t_3)| \le |C(t_1) - C(t_3)| \qquad \text{for all } 0 \le t_1 \le t_2 \le t_3 \le 1, \tag{1.1}$$

that is, if the distance from $C(t)$ to every later point of the curve is nonincreasing in t. Icking, Klein and Langetepe [IKL99] introduced these curves in the analysis of online navigation strategies. Up to the direction of traversal they include the steepest descent curves of quasi-convex functions studied by Manselli and Pucci [MP91], and in the analytic literature they are called *self-contracted curves* [DLS10, DDDL15]; every gradient orbit of a convex function is one. Bounded self-contracted curves in the plane are rectifiable [DLS10]. We always assume C rectifiable and, when convenient, parametrize it by arclength $s \in [0, L]$. Condition (1.1) is invariant under monotone reparametrization.

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The basic metric property of these curves is the *hull inequality*

$$\text{length}(C) \le \text{per}(\text{conv}\, C), \tag{1.2}$$

due to Manselli and Pucci [MP91], who state it as the sharp bound π times the mean width; see also [IKL99], and [AAIKLR01] for generalized self-approaching curves. Since $\text{per}\, K \le \pi \,\text{diam}\, K$ for every planar convex body, (1.2) gives $\text{length}(C) \le \pi \,\text{diam}\, C$. The classical invariant of self-approaching curves is the *detour*, the ratio of length to the distance between the endpoints. Its supremum is $5.3331\ldots$ [IKL99], and for curves with increasing chords it is $2\pi/3$ [Rot94]. We study the corresponding quantity for the diameter,

$$C_{\text{diam}} := \sup \frac{\text{length}(C)}{\text{diam}(C)},$$

the supremum over all self-approaching plane curves of positive diameter.

Theorem 1.1.

$$1 + \frac{2\pi}{3} \;\le\; C_{\text{diam}} \;\le\; \frac{2\pi}{3} + 4\sin\frac{\pi}{12}.$$

Numerically, $3.0943951\ldots \le C_{\text{diam}} \le 3.1296712\ldots$. *The lower bound is the supremum of length over the curves of diameter one that begin on the circle about their last diametral partner (Corollary 7.12); the upper bound is the largest perimeter of a convex set of diameter one with a right-angled corner (Theorem 4.1). The second inequality is strict (Corollary 9.3).*

The upper bound rests on three facts. The first is an exact formula for the perimeter of the hull (Theorem 3.1): $\text{per}(\text{conv}\, C)$ is the integral of $\cos\alpha + \cos\beta$, where α and β are the angles between the tangent and the two edges of the wedge spanned by the rest of the curve. The integrand is at least 1, so $\text{length}(C) = \text{per}(\text{conv}\, C) - \int \sigma\, ds$ with a nonnegative *defect* σ, which vanishes exactly where the rest of the curve fills a right angle with the tangent along one edge. The second is that the whole curve lies in a wedge of opening at most $\pi/2$ at its starting point (Lemma 2.2). The third is the solution of the resulting problem for convex sets: among convex sets of diameter one with a right-angled corner, the trimmed Reuleaux triangle has the largest perimeter, $\frac{2\pi}{3} + 4\sin\frac{\pi}{12}$ (Theorem 4.1). We prove this with an exact dual certificate, a measure μ on directions and a nonnegative function g on the circle for which Cauchy's formula becomes the identity $\text{per}\, K = \int (h_K(u) + h_K(-u))\, d\mu(u) - \int g\, dS_K$ for every such K. Since the bound is attained by a convex body, a better upper bound must use more of the curve: the defect, or the pair condition of Section 9, which the trimmed Reuleaux triangle violates.

For the lower bound, a circular arc of angle $\pi/3$ about the endpoint, followed by a cycloidal cascade of vanishing height on the radius back to the endpoint, has ratio tending to $2 + \pi/3 = 3.0471\ldots$ (Section 5). No curve that winds monotonically about its endpoint before a final cascade does better (Proposition 7.14). Turning back and forth about the endpoint does. In Section 7 we study curves that begin on the circle about their last diametral partner. They lie in a region of perimeter $1 + 2\pi/3 = 3.0943\ldots$, which bounds their length (Proposition 7.3), and the supremum in the class is expressed through the supremal length κ of self-approaching curves between the two corners of a circular segment K with corners of angle $\pi/6$ (Proposition 7.10). The arc-and-cascade curve corresponds to the straight cascade across K, of length 2, while the hull inequality gives $\kappa \le \text{per}\, K = 1 + \pi/3$. We prove $\kappa = 1 + \pi/3$

(Theorem 7.11), so the class bound $1 + 2\pi/3$ is attained in the limit, and it is the lower bound of Theorem 1.1.

The outer arc of K is centred at the starting point of the curve, and a self-approaching curve cannot follow such an arc without slack (Lemma 7.5). It can, however, oscillate along the chords of an inscribed polygon, with all later points on the inner side. Section 6 studies such one-sided oscillations. A cycloidal cascade whose tops are cut off by chords at an angle Θ (Figure 3) has multiplicity $1 + \cos\Theta$, its length per unit of progress, and this is the largest multiplicity of an oscillation whose later points lie in a wedge of angle Θ on one side (Theorem 6.3). The cut angle may change from period to period, and a cut cascade can be followed by a turn of up to Θ, whereas the classical cascade can stand only at the end of a curve (Lemma 7.8). Along a chord of the outer arc we take Θ to be the angle under which the rest of the polygon is seen. Then the length of the oscillation equals the decrease of the perimeter of the hull of the rest of the curve, the defect vanishes, and summing over the chords gives the perimeter of K.

Section 8 describes the polygonal model and the numerical search that preceded this work. A self-approaching polygon with 362 segments in K, certified in exact rational arithmetic, gives an independent finite check that $\kappa > 2$; polygons approach the cut cascades only slowly, with an error of order $1/P$ for P vertices per period. Section 9 shows that the upper bound is strict and that every curve on one side of the line through its starting point and its last diametral partner obeys the lower bound, and collects open problems; the main one is whether $C_{\mathrm{diam}} = 1 + 2\pi/3$.

2. Preliminaries

We write $\mathbf{e}(\theta) = (\cos\theta, \sin\theta)$, $B(x, r)$ for the closed disc, and $\angle(u, v) \in [0, \pi]$ for the angle between nonzero vectors. A *wedge* is a closed convex cone $\{\rho\,\mathbf{e}(\theta) : \rho \geq 0,\ \psi_1 \leq \theta \leq \psi_2\}$ with $0 \leq \psi_2 - \psi_1 < \pi$; its *opening* is $\psi_2 - \psi_1$. For a self-approaching curve $C\colon [0, L] \to \mathbb{R}^2$ let

$$H(s) := \operatorname{conv} C([s, L]) \qquad (0 \leq s \leq L)$$

be the hull of the remaining curve.

Lemma 2.1 (Diameter from the start)**.** $\operatorname{diam} C = \max_t |a - C(t)|$, *where* $a = C(0)$.

Proof. For $s \leq t$, the triple $(0, s, t)$ in (1.1) gives $|C(s) - C(t)| \leq |a - C(t)|$, and every pair of points arises from an ordered pair of parameters. □

Lemma 2.2 (Right angle)**.** *Let* $x = C(s)$ *and* $s \leq t_1 \leq t_2$ *with* $z_i = C(t_i) \neq x$. *Then* $\angle(z_1 - x, z_2 - x) \leq \pi/2$. *Consequently the smallest closed convex cone* $W(s)$ *containing* $C([s, L]) - x$ *is a wedge of opening* $\theta(s) \leq \pi/2$. *If* C *is differentiable at* s *with unit tangent* $T(s)$, *then* $T(s) \in W(s)$ *and* $C([s, L]) \subset \{z : (z - x) \cdot T(s) \geq 0\}$.

Proof. The triple (s, t_1, t_2) in (1.1) gives $|z_1 z_2| \leq |x z_2|$. If $z_1 = z_2$ the angle is 0. If x, z_1, z_2 are collinear with x strictly between z_1 and z_2, then $|z_1 z_2| = |x z_1| + |x z_2| > |x z_2|$, which is impossible; in the remaining collinear case the angle at x is 0. Otherwise the triangle $x z_1 z_2$ is nondegenerate and the longer side lies opposite the larger angle, so $|z_1 z_2| \leq |x z_2|$ gives $\angle x \leq \angle z_1$, and hence $2\angle x \leq \angle x + \angle z_1 < \pi$.

The directions $(z - x)/|z - x|$ of later points, together with their limits, are therefore pairwise at angle at most $\pi/2$. Fix one of them, d_0. All others lie in the closed half-circle of directions within $\pi/2$ of d_0, on which angular distance is the difference of angular coordinates; so the whole set lies in a closed arc of length at

most $\pi/2$, which spans $W(s)$. Since $C(s+h)-x \in W(s)$ for $h>0$ and $W(s)$ is a closed cone, $T(s) \in W(s)$. Finally, every vector of a wedge of opening at most $\pi/2$ makes an angle at most $\pi/2$ with $T(s)$. □

Remark 2.3. For $s=0$, Lemma 2.2 puts the whole curve in a wedge of opening at most $\pi/2$ with apex a. In the extremal family of Section 5, $W(0)$ is a right angle whose edges are the initial tangent and the ray from a through the endpoint f; the initial tangent is perpendicular to af, and the ray from a along it contains no later point.

Lemma 2.4 (Lens)**.** *Let* $\operatorname{diam} C = 1$ *and let* $e = C(t_*)$ *be the last point with* $|a-e|=1$. *Then* $C \subset B(a,1) \cap B(e,1)$. *The vertices* $p_\pm = \frac{1}{2}(a+e) \pm \frac{\sqrt{3}}{2} n$ *of this lens, where* n *is a unit normal of* $e-a$, *satisfy* $|p_+ - p_-| = \sqrt{3} > 1$, *so* C *contains at most one of them.*

Proof. $C \subset B(a,1)$ by Lemma 2.1. For $s \le t_*$ the triple $(0, s, t_*)$ gives $|C(s)-e| \le |a-e| = 1$, and for $s \ge t_*$ the triple $(0, t_*, s)$ gives $|e - C(s)| \le |a - C(s)| \le 1$. □

3. The hull identity

Theorem 3.1 (Hull identity)**.** *Let* $C\colon [0,L] \to \mathbb{R}^2$ *be self-approaching and parametrized by arclength. For almost every* s *the unit tangent* $T(s)$ *exists and lies in* $W(s)$. *Let* $\alpha(s), \beta(s) \ge 0$ *be the angles between* $T(s)$ *and the two edges of* $W(s)$, *so that* $\alpha+\beta = \theta(s) \le \pi/2$. *Then*

$$\operatorname{per}(\operatorname{conv} C) = \int_0^L \big(\cos\alpha(s) + \cos\beta(s)\big)\, ds. \tag{3.1}$$

Corollary 3.2 (Defect)**.**

$$\operatorname{per}(\operatorname{conv} C) - \operatorname{length}(C) = \int_0^L \sigma(s)\, ds, \qquad \sigma := \cos\alpha + \cos\beta - 1 \;\ge\; \cos\theta \;\ge 0. \tag{3.2}$$

In particular $\operatorname{length}(C) \le \operatorname{per}(\operatorname{conv} C) - \int_0^L \cos\theta(s)\, ds$, *which contains* (1.2). *Moreover* $\sigma(s) = 0$ *if and only if* $\{\alpha(s), \beta(s)\} = \{0, \pi/2\}$.

Proof. For fixed $\theta = \alpha+\beta \in [0, \pi/2]$ the function $\alpha \mapsto \cos\alpha + \cos(\theta-\alpha)$ is concave on $[0,\theta]$, so its minimum $1+\cos\theta$ is attained at the endpoints. Hence $\sigma \ge \cos\theta$, and $\sigma = 0$ forces $\theta = \pi/2$ and $\{\alpha,\beta\} = \{0, \pi/2\}$. □

The inequality (1.2) is due to Manselli and Pucci [MP91]; Theorem 3.1 is the corresponding identity, with the defect made explicit.

Definition 3.3. We call σ the *slack*. A parameter s is *saturated* if $\sigma(s) = 0$, that is, if the remaining curve fills a right angle at $C(s)$ with one edge along $T(s)$ and the other along the normal.

Proof of Theorem 3.1. We use Cauchy's formula $\operatorname{per} K = \int_{\mathbb{S}^1} h_K(u)\, d\lambda(u)$ for compact convex $K \subset \mathbb{R}^2$ [Cau41, Sch14], where $h_K(u) = \max_{z \in K} z \cdot u$ and λ is arclength on $\mathbb{S}^1$. For a segment the formula gives twice its length.

For $u \in \mathbb{S}^1$ put $f_u(t) = C(t) \cdot u$ and $M_u(t) = \max_{\tau \in [t,L]} f_u(\tau)$. Then $M_u(t) = h_{H(t)}(u)$, and $M_u(L) = f_u(L)$ with $\int_{\mathbb{S}^1} f_u(L)\, d\lambda(u) = 0$. Hence

$$\operatorname{per}(\operatorname{conv} C) = \int_{\mathbb{S}^1} \big(M_u(0) - M_u(L)\big)\, d\lambda(u). \tag{3.3}$$

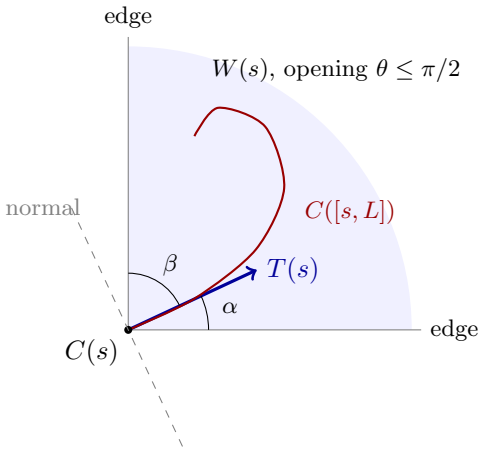


FIGURE 1. The remaining curve lies in a wedge $W(s)$ of opening at most $\pi/2$ (Lemma 2.2). The perimeter of the hull decreases at rate $\cos\alpha + \cos\beta$ (Theorem 3.1), which is at least 1, with equality exactly when $\{\alpha, \beta\} = \{0, \pi/2\}$.

The function M_u is nonincreasing and 1-Lipschitz: for $t < t'$,

$$M_u(t) - M_u(t') \le \max_{\tau\in[t,t']} \big(f_u(\tau) - f_u(t')\big) \le t' - t.$$

Let $A_u = \{t : f_u(t) = M_u(t)\}$; it is closed. On the relatively open set $[0, L] \setminus A_u$ the maximum defining M_u is attained strictly to the right, so M_u is locally constant there. At almost every $t \in A_u$, namely at density points of A_u where both M_u and f_u are differentiable, $M_u'(t) = f_u'(t)$. By absolute continuity,

$$M_u(0) - M_u(L) = \int_0^L \mathbf{1}_{A_u}(t)\big(-C'(t)\cdot u\big)\,dt.$$

The set $\{(t, u) : t \in A_u\}$ is closed, since $M_u(t) - f_u(t)$ is continuous in (t, u), and the integrand is bounded, so Fubini's theorem applied to (3.3) gives

$$\operatorname{per}(\operatorname{conv} C) = \int_0^L \int_{N(t)} \big(-T(t)\cdot u\big)\,d\lambda(u)\,dt,$$

where $N(t) := \{u \in \mathbb{S}^1 : (z - C(t))\cdot u \le 0 \text{ for all } z \in C([t, L])\}$. This set is the polar of $W(t)$ intersected with $\mathbb{S}^1$. If $W(t) = \{\rho\,\mathbf{e}(\psi) : \psi \in [\psi_1, \psi_2]\}$ and $T(t) = \mathbf{e}(\tau)$ with $\tau \in [\psi_1, \psi_2]$, then $N(t)$ is the arc $[\psi_2 + \pi/2, \psi_1 + 3\pi/2]$, and with $\alpha = \tau - \psi_1$, $\beta = \psi_2 - \tau$,

$$\int_{\psi_2+\pi/2}^{\psi_1+3\pi/2} -\cos(\varphi - \tau)\,d\varphi = \sin\Big(\frac{\pi}{2} + \beta\Big) - \sin\Big(\frac{3\pi}{2} - \alpha\Big) = \cos\beta + \cos\alpha. \qquad \square$$

Remark 3.4 (Limits). Length is only lower semicontinuous under uniform convergence. The curves $C_{\pi/3,n}$ of Section 5 have length $2 + \pi/3$ and converge uniformly to an arc followed by a segment, of total length $1 + \pi/3$; in the limit the segment carries multiplicity two. For this reason none of our arguments assumes that a supremum is attained: Corollary 3.2 applies to each curve separately. Supremal ratios are

attained in the relaxed class of pairs (limit curve, limit of lengths): normalized to constant speed on $[0,1]$, curves of diameter one have length at most π, so the Arzelà–Ascoli theorem applies, and (1.1) passes to uniform limits.

4. The upper bound

We write directions in degrees, $u = \mathbf{e}(\theta)$, and use Cauchy's formula $\operatorname{per} K = \int h_K \, d\lambda$. For a compact convex set K let $w_K(\theta) = h_K(\theta) + h_K(\theta + 180^\circ)$ be its width in direction θ, and let S_K be its length measure on $\mathbb{S}^1$; for a polygon, S_K places the length of each edge at its outer normal. Then $h_K + h_K'' = S_K$ in the sense of distributions on the circle [Sch14].

Theorem 4.1 (Largest perimeter with a right-angled corner)**.** *Let K be a compact convex set with $\operatorname{diam} K \le 1$, and suppose that $K \subset z + W$ for a point $z \in K$ and a closed wedge W of opening at most $\pi/2$. Then*

$$\operatorname{per} K \le \frac{2\pi}{3} + 4 \sin \frac{\pi}{12} = 3.1296712828\ldots$$

Equality holds for the trimmed Reuleaux triangle: the Reuleaux triangle of width one with a vertex at z, intersected with the right-angled wedge at z that is symmetric about the axis of the triangle through z.

Proof. Enlarging W keeps the hypotheses, so we may assume $z = 0$ and $W = Q = \{(x,y) : x \ge 0,\ y \ge 0\}$. Then $h := h_K$ vanishes on the arc $N = [180^\circ, 270^\circ]$, because $0 \in K \subset Q$. Put

$$t = \tan 15^\circ = 2 - \sqrt{3}, \qquad s = \tan 7.5^\circ, \qquad c = \frac{t - s}{\sqrt{3}},$$

$$A = \frac{1 + c/4}{\cos 15^\circ}, \qquad B = -\frac{c}{4 \sin 15^\circ}.$$

Let g be the continuous function on $\mathbb{S}^1$ that vanishes on N, satisfies $g(\theta) = g(90^\circ - \theta)$, and is given on $[-90^\circ, 45^\circ]$ by

$$g(\theta) = \begin{cases} \cos(\theta + 82.5^\circ)/\cos 7.5^\circ - 1, & -90^\circ \le \theta \le -75^\circ, \\ 0, & -75^\circ \le \theta \le -45^\circ, \\ c \sin(\theta + 45^\circ), & -45^\circ \le \theta \le -15^\circ, \\ -1 + A\cos\theta + B \sin\theta, & -15^\circ \le \theta \le 15^\circ, \\ 0, & 15^\circ \le \theta \le 45^\circ, \end{cases}$$

and let μ be the measure

$$\mu = \lambda|_{[15^\circ, 75^\circ]} + \lambda|_{[105^\circ, 165^\circ]} + s\,(\delta_{105^\circ} + \delta_{165^\circ}) + c\,\delta_{135^\circ} + (t + c)(\delta_{15^\circ} + \delta_{75^\circ}).$$

We claim that

$$\operatorname{per} K = \int w_K \, d\mu - \int g \, dS_K. \tag{4.1}$$

Let $\tilde\mu(X) = \mu(X) + \mu(X - 180^\circ)$, so that $\int w_K \, d\mu = \int h \, d\tilde\mu$. On the open arcs of the list, $g + g''$ equals -1 on $(-90^\circ, -75^\circ)$ and on $(-15^\circ, 15^\circ)$, where $\tilde\mu$ has no mass, and 0 on the three other arcs, where $\tilde\mu$ has density one; by the symmetry the same holds on $[45^\circ, 180^\circ]$. At the breakpoints g is continuous, and g' jumps by s at -75° and at -15°, by c at -45°, and by $t + c$ at 15°, and $g'(45^\circ) = 0$. These jumps are

the atoms of $\tilde{\mu}$ at these directions, and by symmetry at 165°, 105°, 135° and 75°. Hence $\tilde{\mu} - (g + g'') = \lambda$ as measures on the open arc $\mathbb{S}^1 \setminus N$. Since h vanishes on N,

$$\operatorname{per} K = \int_{\mathbb{S}^1 \setminus N} h \, d\lambda = \int h \, d\tilde{\mu} - \int h \, d(g + g'') = \int w_K \, d\mu - \int g \, dS_K,$$

where $\int h \, d(g + g'') = \int g \, d(h + h'')$ follows from two integrations by parts on the circle: h and g are Lipschitz, h' and g' have bounded variation, and in each step one of the two factors is continuous. This proves (4.1).

On $[-15°, 15°]$, $g = -1 + \rho \cos(\theta - \theta_0)$ with $\rho > 0$ and $|\theta - \theta_0| < 90°$ is concave, with values $c/2$ and 0 at the ends, so $g \geq 0$ there, and $g \geq 0$ on the other pieces is clear. Since $w_K \leq \operatorname{diam} K \leq 1$ and $S_K \geq 0$, (4.1) gives $\operatorname{per} K \leq \mu(\mathbb{S}^1) = \frac{2\pi}{3} + 2s + c + 2(t + c)$. With $3c = \sqrt{3}(t - s)$, $(2 + \sqrt{3})\, t = 1$ and $1 + \tan a \tan \frac{a}{2} = \sec a$ this equals $\frac{2\pi}{3} + 1 + ts = \frac{2\pi}{3} + \sec 15° = \frac{2\pi}{3} + 4 \sin 15°$.

In these coordinates the trimmed Reuleaux triangle K_* is the intersection of Q with the Reuleaux triangle on the vertices 0, $\mathbf{e}(15°)$ and $\mathbf{e}(75°)$. Its boundary consists of the arc of radius one about 0 from $\mathbf{e}(15°)$ to $\mathbf{e}(75°)$, of length $\pi/3$; the arcs of radius one about $\mathbf{e}(75°)$ and about $\mathbf{e}(15°)$ from these points to the axes, of length $\pi/6$ each; and two segments of length $2 \sin 15°$ on the axes. So $\operatorname{per} K_* = \frac{2\pi}{3} + 4 \sin 15°$. □

Equality in Theorem 4.1 holds only for K_* (in the normalized position). Indeed, equality forces $w_K = 1$ on the support of μ and $S_K = 0$ on the open set $\{g > 0\}$. Then $h_K = 1$ on $[15°, 75°]$, so K contains the arc of radius one about 0 between $\mathbf{e}(15°)$ and $\mathbf{e}(75°)$; since $\operatorname{diam} K \leq 1$, K therefore lies in the discs of radius one about 0, $\mathbf{e}(15°)$ and $\mathbf{e}(75°)$, and it lies in Q. The intersection of these four sets is K_*, and a convex subset of K_* with the same perimeter is K_* itself. The pair (μ, g) is the continuous limit of the optimal dual solutions of the linear programs over support functions sampled at n directions. These give 3.1297245 for $n = 360$ and 3.1296721 for $n = 2880$, and decrease to the value of the theorem like n^{-2}.

Theorem 4.2. *Every self-approaching curve C satisfies*

$$\operatorname{length}(C) \leq \Big(\frac{2\pi}{3} + 4 \sin \frac{\pi}{12} \Big) \operatorname{diam} C.$$

Proof. Normalize $\operatorname{diam} C = 1$, let $a = C(0)$, and let $K = \operatorname{conv} C$. Then $\operatorname{diam} K = 1$ and $a \in K$, and $K \subset a + W(0)$ for a wedge $W(0)$ of opening at most $\pi/2$ by Lemma 2.2. By Corollary 3.2 and Theorem 4.1, $\operatorname{length} C \leq \operatorname{per} K \leq \frac{2\pi}{3} + 4 \sin \frac{\pi}{12}$. □

Remark 4.3 (Limits of the hull method). Theorem 4.2 uses only three properties of $K = \operatorname{conv} C$: its diameter, the point $a \in K$, and the right angle at a. The trimmed Reuleaux triangle has all three, so these properties cannot give a better bound. A further improvement must use more of the structure of the curve: the defect σ in (3.2), or the pair condition of Proposition 9.2, which the trimmed Reuleaux triangle violates (Corollary 9.3). The curves that approach the lower bound (Corollary 7.12) have as their hull the half-lens bounded by the segment from a to its last diametral partner e and by two arcs of angle $\pi/3$ about e and about a, of perimeter $1 + 2\pi/3$, and their defect tends to zero.

5. A lower bound: arcs and cascades

5.1. **The cycloidal cascade.** Fix $r > 0$ and write $c(\varphi) = r(\varphi - \sin\varphi, 1 - \cos\varphi)$ and $\tilde{c}(\varphi) = r(\varphi + \sin\varphi, \cos\varphi - 1)$. The curve $\tilde{c}$ is the evolute of c: it is c shifted by half a period and by $2r$ downwards. For $k \geq 0$ let

$$D_k = \{c(\varphi) - (0, 4kr) : \varphi : \pi \to 2\pi\}, \qquad E_k = \{\tilde{c}(\varphi) - (0, 4kr) : \varphi : 2\pi \to \pi\},$$

traversed in the indicated directions. Each is a half-cycloid of length $4r$. D_k runs from $(\pi r, 2r - 4kr)$ to $(2\pi r, -4kr)$, and E_k from $(2\pi r, -4kr)$ to $(\pi r, -2r - 4kr)$. Let Γ_n be the concatenation $D_0 E_0 D_1 E_1 \cdots$ of n pieces, with n even.

Lemma 5.1 (Cycloidal cascade [IKL97, IKL99])**.** *Γ_n is self-approaching, has length $4nr$, and lies in the rectangle $[\pi r, 2\pi r] \times [2r - 2nr, 2r]$ of width πr and height $2nr$. It starts at $(\pi r, 2r)$ and ends at $(\pi r, 2r - 2nr)$, on the same long side. Placed on a segment of length w with n pieces ($r = w/2n$), it has length $2w$ and lies in a rectangle of height $\pi w/(2n)$ over that segment.*

Proof. Each piece has length $\int_\pi^{2\pi} 2r\sin(\varphi/2)\, d\varphi = 4r$. The glide reflection $G(x, y) = (3\pi r - x,\, y - 2r)$ maps D_k onto E_k and E_k onto D_{k+1}, preserving the direction of traversal. It therefore suffices to show that for $x = c(\varphi)$ with $\varphi \in (\pi, 2\pi)$ and every later point z of the infinite cascade, $(z - x) \cdot c'(\varphi) \geq 0$; then $|\Gamma(t) - z|^2$ is nonincreasing up to the time of z, which is (1.1). Write $c'(\varphi) = 2r\sin\frac{\varphi}{2}\,(\sin\psi, \cos\psi)$ with $\psi = \varphi/2 \in (\pi/2, \pi)$.

Later points of D_0. The tangent of D_0 turns monotonically through $\pi/2$, so every chord to a later point makes an angle at most $\pi/2$ with $c'(\varphi)$.

Points of E_0. The normal line of c at x is tangent to the evolute $\tilde{c}$ at $\tilde{c}(\varphi) \in E_0$. Since E_0 is a convex arc, it lies on one side of this tangent line, and it is the correct side, because for the first point $B = (2\pi r, 0)$ of E_0 we have $(B - x) \cdot c'(\varphi) = r^2(1 - \cos\varphi)(2\pi - \varphi) \geq 0$.

All later pieces. They lie in $\{\pi r \leq z_1 \leq 2\pi r,\ z_2 \leq -2r\}$. The function $Q(z) = (z - x) \cdot (\sin\psi, \cos\psi)$ increases in z_1 and decreases in z_2, so its minimum over this region is attained at $z = (\pi r, -2r)$. With $t = \psi - \pi/2 \in (0, \pi/2)$ a direct computation gives $Q(\pi r, -2r) = 2r(\sin t - t\cos t) \geq 0$.

The remaining statements are read off from the formulas. □

A cascade can replace the last segment of any self-approaching polygon whose inequalities hold with a margin. For points $v_0, \ldots, v_N$ put

$$\Lambda(v) = \sum_{i<N} |v_i - v_{i-1}| + 2|v_N - v_{N-1}|. \tag{5.1}$$

Lemma 5.2 (Terminal cascade)**.** *Let $\Omega \subset \mathbb{R}^2$ be closed and convex, and let $v_0, \ldots, v_N \in \Omega$ be distinct points with*

$$(v_j - v_i) \cdot (v_i - v_{i-1}) \ \geq\ \eta\,|v_j - v_i|\,|v_i - v_{i-1}| \qquad (1 \leq i < j \leq N) \tag{5.2}$$

for some $\eta > 0$, and suppose that the open segment (v_{N-1}, v_N) lies in the interior of Ω. Then for every $\varepsilon > 0$ there is a self-approaching curve in Ω from v_0 to v_N of length at least $\Lambda(v) - 2\varepsilon$.

Proof. For a curve C and a later point z, $s \mapsto |C(s) - z|^2$ is absolutely continuous with derivative $2(C(s) - z) \cdot C'(s)$. So C is self-approaching if, for almost every s, every later point z satisfies $(z - C(s)) \cdot C'(s) \geq 0$; we call this the half-plane criterion.

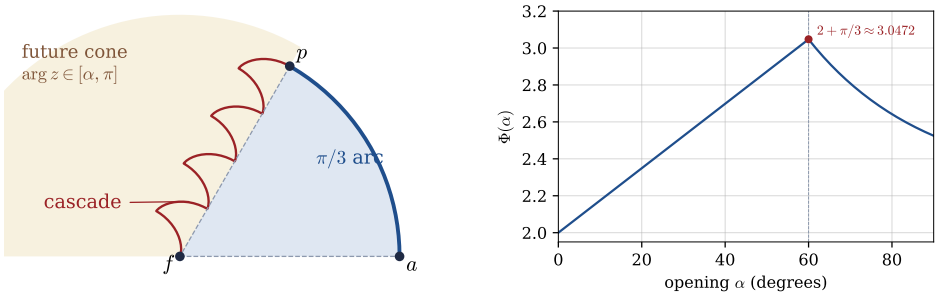


FIGURE 2. Left: $C_{\pi/3,8}$, computed from the explicit formulas, with the arc about f followed by the cascade on the future-cone side of pf. Right: the limiting ratio $\Phi(\alpha)$ of the family (Proposition 5.6).

Let $w = |v_N - v_{N-1}|$, $u = (v_N - v_{N-1})/w$, $0 < \varepsilon < w/2$, $x_0 = v_{N-1} + \varepsilon u$ and $x_1 = v_N - \varepsilon u$. Replace the last segment by $[v_{N-1}, x_0]$, the cascade Γ of Lemma 5.1 with n pieces on $[x_0, x_1]$, and $[x_1, v_N]$. The length is $\Lambda(v) - 2\varepsilon$. The cascade lies within $h = \pi(w - 2\varepsilon)/(2n)$ of $[x_0, x_1]$, a compact subset of the interior of Ω, so it lies in Ω for large n; its points project onto the line of u inside $[x_0, x_1]$.

No vertex v_i with $i < N - 1$ lies on (v_{N-1}, v_N): a self-approaching curve that returns to a point is constant in between, and the vertices are distinct. Hence $m = \min_{i<N} \operatorname{dist}(v_i, [x_0, x_1]) > 0$. Let u_i be the unit direction of the i-th segment, $i < N$. The function $f_i(z) = (z - v_i) \cdot u_i - \eta|z - v_i|$ is concave and nonnegative at v_{N-1} and v_N by (5.2), so it is nonnegative on $[v_{N-1}, v_N]$, and $(z' - v_i) \cdot u_i \geq \eta m$ for $z' \in [x_0, x_1]$. A cascade point z lies within h of such a z', so $(z - v_i) \cdot u_i \geq \eta m - h \geq 0$ once $h \leq \eta m$. For x on the i-th segment, $(z - x) \cdot u_i \geq (z - v_i) \cdot u_i$.

We check the half-plane criterion. For x on a segment $i < N$, later polygon points satisfy it by (5.2), and later cascade points by the last paragraph. For x on $[v_{N-1}, x_0]$, every later point projects onto the line of u beyond x. For x on the cascade, later cascade points satisfy it by Lemma 5.1, and points $z = x_1 + \tau u$ of the final straight piece satisfy it as in the proof of Lemma 7.4(i): on the last half-cycloid, $(z - x) \cdot T(x) = (x_1 - x) \cdot T(x) + \tau\, u \cdot T(x) \geq 0$, and for earlier pieces the bound $Q \geq 0$ from the proof of Lemma 5.1 applies. Points of the final straight piece are followed only by points ahead of them. □

5.2. **The curves $C_{\alpha,n}$.** Place $f = (0, 0)$ and $a = (1, 0)$. For $\alpha \in (0, \pi/2]$ let $p = \mathbf{e}(\alpha)$. The curve $C_{\alpha,n}$ consists of the unit arc $\theta \mapsto \mathbf{e}(\theta)$, $0 \leq \theta \leq \alpha$, from a to p, followed by the cascade of Lemma 5.1 with n pieces on the segment from p to f. The rectangle of the cascade lies on the side of pf where the argument exceeds α, along the unit normal $\mathbf{e}(\alpha + \pi/2)$ (Figure 2).

Lemma 5.3 (Arc)**.** *A circular arc of angle at most π is self-approaching.*

Proof. On a circle of radius ρ, chords have length $2\rho \sin(\Delta/2)$, where $\Delta \in [0, \pi]$ is the subtended angle, and this is increasing in Δ. □

Lemma 5.4 (Future cone)**.** *Let $z = f + \rho\, \mathbf{e}(\psi)$ with $\rho > 0$. The distance from $\mathbf{e}(\theta)$ to z is nonincreasing for $\theta \in [0, \alpha]$ if and only if $\psi \in [\alpha, \pi]$ modulo 2π. The point $z = f$ is at constant distance* 1.

Proof. $\frac{d}{d\theta}|\mathbf{e}(\theta) - z|^2 = -2\rho\sin(\psi - \theta)$. This is ≤ 0 for all $\theta \in [0, \alpha]$ if and only if $\psi \in \bigcap_{\theta\in[0,\alpha]}[\theta, \theta + \pi] = [\alpha, \pi]$. □

Theorem 5.5. *For $\alpha \in (0, \pi/2]$ and even $n \ge 2$, $C_{\alpha,n}$ is self-approaching, has length $\alpha + 2$, and satisfies* $\operatorname{diam} C_{\alpha,n} \le \max(1, 2\sin\frac{\alpha}{2}) + \frac{\pi}{2n}$. *In particular*

$$\frac{\operatorname{length}(C_{\pi/3,n})}{\operatorname{diam}(C_{\pi/3,n})} \ge \frac{2 + \pi/3}{1 + \pi/2n},$$

and therefore $C_{\mathrm{diam}} \ge 2 + \pi/3$.

Proof. The cascade has length $2|pf| = 2$. A point of its rectangle has the form $(1 - \tau)p + y\,\mathbf{e}(\alpha + \pi/2)$ with $\tau \in [0, 1]$ and $y \in [0, \pi/2n]$. Its argument lies in $[\alpha, \alpha + \pi/2] \subset [\alpha, \pi]$, so by Lemma 5.4 the distance from the arc to every point of the tail is nonincreasing along the arc. Triples inside the arc or inside the tail are covered by Lemmas 5.3 and 5.1. If t_1 lies on the arc and $t_2 \le t_3$ lie on the tail, then $|C(t_2) - C(t_3)| \le |p - C(t_3)| \le |C(t_1) - C(t_3)|$. Every point of the tail lies within $\pi/2n$ of $[p, f]$, and $[p, f] \subset B(a, \max(1, |ap|))$ by convexity. Hence $\max_t |a - C(t)| \le \max(1, 2\sin\frac{\alpha}{2}) + \frac{\pi}{2n}$, and Lemma 2.1 gives the diameter. □

The limiting ratio of the family is $\Phi(\alpha) = (\alpha + 2)/\max(1, 2\sin\frac{\alpha}{2})$ on $(0, \pi/2]$. Section 7 uses the same formula on $(0, \pi)$.

Proposition 5.6 (Best opening within the family)**.** *Φ is strictly increasing on $(0, \pi/3]$ and strictly decreasing on $[\pi/3, \pi/2]$. Its maximum on $(0, \pi/2]$ is $\Phi(\pi/3) = 2 + \pi/3$.*

Proof. On $(0, \pi/3]$, $\Phi(\alpha) = \alpha + 2$. On $[\pi/3, \pi/2]$ the logarithmic derivative of $(\alpha + 2)/(2\sin\frac{\alpha}{2})$ has the sign of $\tan u - (u + 1)$ with $u = \alpha/2 \in [\pi/6, \pi/4]$. Since $\tan u - u$ is increasing, this is at most $\tan\frac{\pi}{4} - \frac{\pi}{4} - 1 = -\frac{\pi}{4} < 0$. □

Remark 5.7. Proposition 5.6 compares members of a single family, each rescaled to diameter one; it is not an upper bound for C_{diam}. For $\alpha > \pi/3$ the decrease of Φ only reflects the growth of the diameter: inside $B(a, 1)$, an arc of radius one about f that starts at a cannot be continued beyond the angle $\pi/3$.

6. One-sided oscillations and cut cascades

In this section $e_1 = (1, 0)$, and for $\Theta \in [0, \pi/2)$ we write

$$V_\Theta = \{\rho\,\mathbf{e}(\alpha) : \rho \ge 0,\ 0 \le \alpha \le \Theta\}, \qquad \mathcal{T}_\Theta = \{\mathbf{e}(\gamma) : \Theta - \tfrac{\pi}{2} \le \gamma \le \tfrac{\pi}{2}\}.$$

A unit vector T satisfies $T \cdot v \ge 0$ for all $v \in V_\Theta$ if and only if $T \in \mathcal{T}_\Theta$. By the half-plane criterion of the proof of Lemma 5.2, a curve whose unit tangents lie in $\mathcal{T}_\Theta$ approaches every point that is far away in a direction of V_Θ. This is the situation of an oscillation of vanishing height along e_1 whose later points lie in a wedge of angle Θ on one side of its direction.

Definition 6.1 (Cut cascades)**.** Let $R > 0$ and

$$U(\psi) = R\Big(\tfrac{1}{2}\cos^2\psi,\ \tfrac{\pi}{4} - \tfrac{\psi}{2} + \tfrac{1}{4}\sin 2\psi\Big), \qquad D(\varphi) = R\Big(1 - \tfrac{1}{2}\sin^2\varphi,\ -\tfrac{\varphi}{2} + \tfrac{1}{4}\sin 2\varphi\Big).$$

For $\Theta \in [0, \pi/2)$ the *period cut at* Θ runs along U from $U(\pi/2) = 0$ to $U(\Theta)$, then along the chord from $U(\Theta)$ to $D(\Theta - \frac{\pi}{2})$, then along D to $D(0) = Re_1$. For

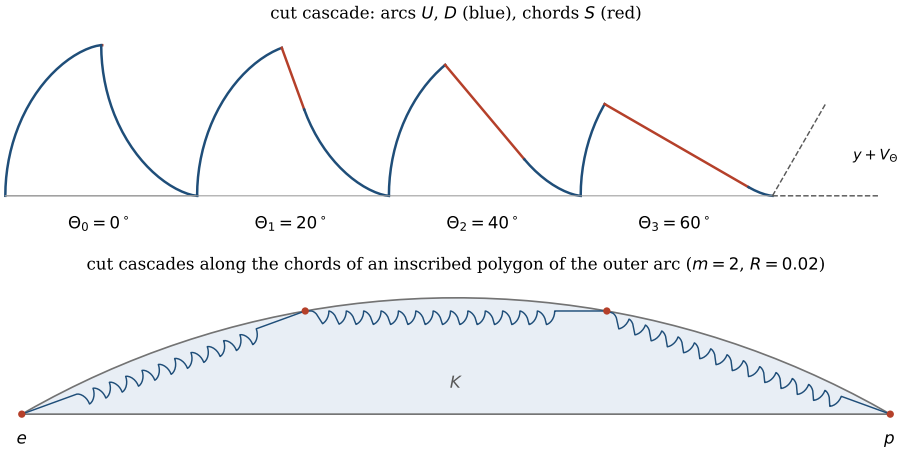


FIGURE 3. Top: a cut cascade with cut angles 0°, 20°, 40° and 60°; the first period is a period of the cycloidal cascade. Every point of a period cut at Θ approaches the whole wedge $y + V_\Theta$ at the end y of its period. Bottom: the curve of Theorem 7.11 for $m = 2$ and $R = 0.02$ in the cap K, with the chord ep horizontal: cut cascades along the three chords of an inscribed polygon of the outer arc, bulging to the inner side.

$\Theta_0, \dots, \Theta_{n-1} \in [0, \pi/2)$ the *cut cascade* $Z = Z(R; \Theta_0, \dots, \Theta_{n-1})$ is the concatenation of the periods cut at Θ_k and translated by kRe_1, $k = 0, \dots, n-1$. Its k-th period ends at $y_k = (k+1)Re_1$.

Lemma 6.2. (i) *$U'(\psi) = -R\sin\psi\,\mathbf{e}(\psi)$ and $D'(\varphi) = -R\sin\varphi\,\mathbf{e}(\varphi)$. So U, traversed with ψ decreasing, and D, traversed with φ increasing, have the unit tangents $\mathbf{e}(\psi)$ and $\mathbf{e}(\varphi)$. Moreover*

$$D(\psi - \tfrac{\pi}{2}) - U(\psi) = R\sin\psi\,\mathbf{e}(\psi - \tfrac{\pi}{2}),$$
$$U(\varphi + \tfrac{\pi}{2}) + Re_1 - D(\varphi) = R\big(e_1 - \cos\varphi\,\mathbf{e}(\varphi)\big). \tag{6.1}$$

(ii) *The period cut at Θ has length $R(1+\cos\Theta)$ and lies in $[0,R]\times[0,\frac{\pi}{4}R]$, its unit tangents lie in $\mathcal{T}_\Theta$, and its chord has direction $\mathbf{e}(\Theta - \frac{\pi}{2})$ and length $R\sin\Theta$. For $\Theta = 0$ the chord is a point, and the period consists of two half-cycloids of the cascade of Lemma 5.1.*

(iii) *Every cut cascade is self-approaching.*

(iv) *If x lies on the k-th period of Z, then $(z-x)\cdot T(x) \geq 0$ for every $z \in y_k + V_{\Theta_k}$.*

Proof. (i) and (ii) are direct computations. The chord is the case $\psi = \Theta$ of the first identity in (6.1), and the length is $R\cos\Theta + R\sin\Theta + R(1-\sin\Theta)$. The radii of curvature $R\sin\psi$ and $R|\sin\varphi|$ are proportional to the sine of the angle between the tangent and e_1, so U and D are arcs of cycloids.

(iii) Let x lie on the k-th period, with unit tangent $\mathbf{e}(\gamma)$, $\gamma \in [\Theta_k - \frac{\pi}{2}, \frac{\pi}{2}]$, and put $f(z) = (z-x)\cdot\mathbf{e}(\gamma)$. Along the rest of Z the derivative of f is $\cos(\gamma' - \gamma)$, where $\mathbf{e}(\gamma')$ is the tangent there. On period j the angle γ' decreases from $\pi/2$ to Θ_j along U, equals $\Theta_j - \frac{\pi}{2}$ on the chord, and increases from $\Theta_j - \frac{\pi}{2}$ to 0 along D.

Since $|\gamma' - \gamma| < \pi$, f decreases exactly where $|\gamma' - \gamma| > \pi/2$. Right after x it does not decrease, and $f(x) = 0$, so it suffices to show $f \geq 0$ at the local minima of f on the rest of Z. The arcs U and D of period j are pieces of the fixed arcs $U + jRe_1$ and $D + jRe_1$, cut at Θ_j.

$x = U(\psi) + kRe_1$, $\gamma = \psi \geq \Theta_k$. Here f decreases only where $\gamma' < \psi - \frac{\pi}{2}$: on period $j \geq k$ this happens on the chord and on D up to the angle $\psi - \frac{\pi}{2}$ if $\Theta_j < \psi$, and nowhere otherwise. The local minima are the points $D(\psi - \frac{\pi}{2}) + jRe_1$ with $j \geq k$ and $\Theta_j \leq \psi$, and there, by (6.1),

$$f = \big(D(\psi - \tfrac{\pi}{2}) - U(\psi)\big) \cdot \mathbf{e}(\psi) + (j-k)R\, e_1 \cdot \mathbf{e}(\psi) = (j-k)R\cos\psi \geq 0.$$

$x = D(\varphi) + kRe_1$, $\gamma = \varphi \leq 0$. Here f increases on the rest of period k, and on period $j > k$ it decreases only along U while $\gamma' > \varphi + \frac{\pi}{2}$. Its minimum on period j is attained at $U(\max(\Theta_j, \varphi + \frac{\pi}{2})) + jRe_1$, and it is at least the value at $U(\varphi + \frac{\pi}{2}) + jRe_1$, where f stops decreasing along the uncut arc. By (6.1) that value is $(j-k-1)R\cos\varphi \geq 0$.

$x = U(\Theta_k) + \tau\, \mathbf{e}(\Theta_k - \frac{\pi}{2}) + kRe_1$ *on the chord,* $0 \leq \tau \leq R\sin\Theta_k$. As before f increases on the rest of period k, and on period $j > k$ its minimum is at least its value at $U(\Theta_k) + jRe_1$, which is $(j-k)R\, e_1 \cdot \mathbf{e}(\Theta_k - \frac{\pi}{2}) - \tau = (j-k)R\sin\Theta_k - \tau \geq 0$.

By the half-plane criterion, Z is self-approaching.

(iv) The end y_k of period k is a later point of x, so $(y_k - x) \cdot T(x) \geq 0$ by the proof of (iii). For $z = y_k + \rho\, \mathbf{e}(\alpha)$ with $0 \leq \alpha \leq \Theta_k$ also $\rho\, \mathbf{e}(\alpha) \cdot T(x) \geq 0$, because $T(x) \in \mathcal{T}_{\Theta_k}$ by (ii). □

By (6.1) the normal line at $U(\psi)$ passes through the later point $D(\psi - \frac{\pi}{2})$ of the same period, and the normal line at $D(\varphi)$ through the point $U(\varphi + \frac{\pi}{2})$ of the next period. These points play the role of the contact points of Lemma 7.6, and they run over later pieces of the curve. Cutting the tops of the cycloidal cascade by chords keeps them, because the arcs U and D do not depend on Θ.

Theorem 6.3 (One-sided oscillations)**.** *Let $\Theta \in [0, \pi/2)$, and let $Z\colon \mathbb{R} \to \mathbb{R}^2$ be parametrized by arclength, self-approaching on every compact interval, with unit tangents in $\mathcal{T}_\Theta$, and periodic in the sense that $Z(s + \tau) = Z(s) + \ell e_1$ for some $\tau, \ell > 0$ and all s. Then $\tau \leq (1 + \cos\Theta)\,\ell$. The cut cascade with all cut angles equal to Θ, continued periodically, attains equality.*

Proof. The second statement is Lemma 6.2(ii),(iii) with $\ell = R$. For the first, let $n \geq 1$ and $M > 0$, put $x_0 = Z(0)$, $x_n = Z(n\tau) = x_0 + n\ell e_1$ and $F = x_n + M\mathbf{e}(\Theta)$, and let C be $Z|_{[0,n\tau]}$ followed by the segment $[x_n, F]$. Then C is self-approaching: for $x = Z(s)$ and $z = x_n + \rho\, \mathbf{e}(\Theta)$ we have $(z-x)\cdot T(s) = (x_n - x)\cdot T(s) + \rho\, \mathbf{e}(\Theta)\cdot T(s) \geq 0$, and points of the segment are followed only by points ahead of them. The wedges $W(s)$ of C for $s \geq n\tau$ are those of the final segment, so Theorem 3.1, applied to C and to the segment, and $\cos\alpha + \cos\beta \geq 1$ give

$$n\tau \leq \int_0^{n\tau} (\cos\alpha + \cos\beta)\, ds = \operatorname{per}(\operatorname{conv} C) - 2M.$$

By periodicity $Z([0, n\tau]) \subset [x_0, x_n] + B(0, r)$ for an r independent of n. Since $h_{X+B(0,r)} = h_X + r$, Cauchy's formula gives

$$\operatorname{per}(\operatorname{conv} C) \leq \operatorname{per}\operatorname{conv}\{x_0, x_n, F\} + 2\pi r = n\ell + M + |F - x_0| + 2\pi r,$$

and $|F - x_0| = |n\ell e_1 + M\mathbf{e}(\Theta)| \leq M + n\ell\cos\Theta + n^2\ell^2/(2M)$. Letting $M \to \infty$ gives $n\tau \leq n\ell(1 + \cos\Theta) + 2\pi r$ for every n, and the claim follows. □

The same argument, with later points far away in a wedge whose edges make angles θ_1 and θ_2 with e_1 on the two sides, bounds the multiplicity by $\cos\theta_1 + \cos\theta_2$; the zig-zag of Remark 7.9 attains this for $\theta_1 = \theta_2 = \pi/4$. Theorem 6.3 is the one-sided case $\theta_1 = 0$, $\theta_2 = \Theta$, and Lemma 6.2(iv) shows that the cut cascade does not need the later points to be far away: they may start at the end of any period.

7. Curves that begin on the circle about their last diametral partner

Throughout this section $\operatorname{diam} C = 1$, $a = C(0)$, and $e = C(t_*)$ is the last point with $|a - e| = 1$ (Lemma 2.4). We use coordinates with $e = 0$ and $a = \mathbf{e}(0) = (1, 0)$. The circle $\partial B(e, 1)$ meets $B(a, 1)$ in the arc $\{\mathbf{e}(\theta) : |\theta| \le \pi/3\}$.

7.1. The class $\mathcal{A}_\alpha$ and its region.

Definition 7.1. For $\alpha \in (0, \pi/3]$ let $\mathcal{A}_\alpha$ be the class of curves of diameter one whose initial piece of length α is the arc $\theta \mapsto \mathbf{e}(\theta)$, $0 \le \theta \le \alpha$, about their last diametral partner e (after a reflection). Put $q = \mathbf{e}(\alpha)$ and $p = \mathbf{e}(\pi/3)$.

The curves $C_{\pi/3,n}$ of Section 5 are not in this class, because their farthest point from a lies on the cascade. However, the arc from a to q followed by the curve $\Gamma_{\varepsilon,n}$ of Lemma 7.4 belongs to $\mathcal{A}_\alpha$, and its length tends to $2 + \pi/3$ for every α.

Lemma 7.2 (Where the rest of the curve lies)**.** *Let $C \in \mathcal{A}_\alpha$. Then $C([\alpha, L]) \subset R_\alpha$, where*

$$R_\alpha := \{\rho\,\mathbf{e}(\theta) : \rho \ge 0,\ \alpha \le \theta \le \pi\} \cap B(a, 1) \cap B(e, 1),$$

and the curve ends at e: no point after t_* differs from e. Moreover $R_\alpha = S_\alpha \cup K$, where

$$S_\alpha = \Big\{\rho\,\mathbf{e}(\theta) : 0 \le \rho \le 1,\ \alpha \le \theta \le \frac{\pi}{3}\Big\},$$
$$K = \Big\{\rho\,\mathbf{e}(\theta) : \frac{\pi}{3} \le \theta \le \frac{\pi}{2},\ 0 \le \rho \le 2\cos\theta\Big\},$$

and $\operatorname{per} R_\alpha = 1 + \frac{2\pi}{3} - \alpha$. The cap $K = R_{\pi/3}$ is the circular segment of $B(a, 1)$ cut off by the chord ep, with corners of angle $\pi/6$ at p and at e (Figure 4).

Proof. By Lemma 5.4, with $f = e$, every later point other than e has argument in $[\alpha, \pi]$, and Lemma 2.4 supplies $B(a, 1) \cap B(e, 1)$. For $\rho > 0$, $\rho\,\mathbf{e}(\theta) \in B(a, 1)$ if and only if $\rho \le 2\cos\theta$, which forces $\theta \le \pi/2$; moreover $2\cos\theta \ge 1$ exactly when $\theta \le \pi/3$. This gives the decomposition. The boundary of R_α consists of $[e, q]$ (length 1), the arc of $\partial B(e, 1)$ from q to p (length $\pi/3 - \alpha$), and the arc of $\partial B(a, 1)$ from p to e (length $\pi/3$).

The point e is visited only at t_*: if $C(s) = e$ for some $s < t_*$, the triples (s, t, t_*) would force C to be constant on $[s, t_*]$. Suppose that $z = C(t) \ne e$ for some $t > t_*$. Then $z = \rho\,\mathbf{e}(\theta)$ with $\rho > 0$ and $\theta \in [\alpha, \pi/2]$. Points $x = C(s)$ with $s < t_*$ close to t_* have the form $\rho_x \mathbf{e}(\theta_x)$ with $\theta_x \in [\alpha, \pi/2]$ and $\rho_x > 0$ small. Since $|\theta_x - \theta| < \pi/2$,

$$|x - z|^2 - |e - z|^2 = \rho_x^2 - 2\rho_x\rho\cos(\theta_x - \theta) < 0,$$

which contradicts (1.1) for the triple (s, t_*, t). □

Proposition 7.3 (Hull bound in the class)**.** *Every $C \in \mathcal{A}_\alpha$ satisfies* $\operatorname{length}(C) \le 1 + \frac{2\pi}{3} = 3.0943\ldots$.

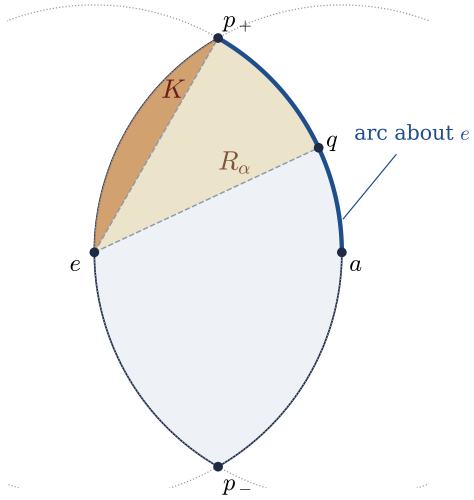


FIGURE 4. The lens $B(a,1) \cap B(e,1)$ in the frame of Section 7. A curve of $\mathcal{A}_\alpha$ leaves the circle about e at $q = \mathbf{e}(\alpha)$. The rest of the curve lies in $R_\alpha = S_\alpha \cup K$ (light); the cap K is darker. For $\alpha = \pi/3$, $R_\alpha = K$.

Proof. R_α is convex, as the intersection of a convex cone of opening $\pi - \alpha < \pi$ with two discs, and perimeter is monotone under inclusion of convex sets. By Lemma 7.2 and Corollary 3.2, $\operatorname{length}(C) = \alpha + \operatorname{length}(C|_{[\alpha,L]}) \le \alpha + \operatorname{per} R_\alpha = 1 + \frac{2\pi}{3}$. □

The value $2 + \pi/3 = 3.0471\ldots$ of the arc-and-cascade construction lies 0.047 below this bound. The difference is exactly $\operatorname{per} K - 2$, the perimeter of the cap minus twice its chord, and Corollary 7.12 shows that the bound is attained in the limit.

Lemma 7.4 (Competitors)**.** *Fix $\alpha \in (0, \pi/3]$ and a small $\varepsilon > 0$, and let $\nu = \mathbf{e}(5\pi/6)$ be the unit normal of the radius $[e,p]$ on the side of larger argument.*

(i) *Let $\Gamma_{\varepsilon,n}$ consist of four pieces: the radial segment from q to $(1-\varepsilon)q$; the arc of radius $1-\varepsilon$ about e from $(1-\varepsilon)q$ to $(1-\varepsilon)p$; the cascade of Lemma 5.1 with n pieces on the segment from $(1-\varepsilon)p$ to εp, on the side of ν; and the segment from εp to e. For $n \ge n_0(\varepsilon)$ this is a self-approaching curve in R_α from q to e, of length*

$$2\varepsilon + (1-\varepsilon)\Big(\frac{\pi}{3} - \alpha\Big) + 2(1-2\varepsilon) \xrightarrow[\varepsilon\to 0]{} 2 + \frac{\pi}{3} - \alpha.$$

For $\alpha = \pi/3$ the arc is empty, and $\Gamma_{\varepsilon,n}$ is the straight cascade in K, of length $2 - 2\varepsilon$.

(ii) *In K, take the arc of $\partial B(a,1)$ from p through an angle $\beta \in [0, \pi/3]$, followed by a cascade on the chord from its endpoint to e, modified as in (i) near both ends of the chord. These curves lie in K, and their lengths tend to $g(\beta) = \beta + 4\sin(\frac{\pi}{6} - \frac{\beta}{2})$, which decreases strictly from $g(0) = 2$ to $g(\pi/3) = \pi/3$.*

Proof. (i) *Containment.* A point of the rectangle of the cascade is $z = tp + y\nu$ with $t \in [\varepsilon, 1-\varepsilon]$ and $0 \le y \le h = \pi(1-2\varepsilon)/(2n)$. Then $|z-a|^2 = 1-t(1-t)+y^2+\sqrt{3}\,y$, which is at most 1 once $y^2 + \sqrt{3}\,y \le \varepsilon(1-\varepsilon)$; this holds for $n \ge n_0(\varepsilon)$. Then $z \in K$: the argument of z is at least $\pi/3$ because $y \ge 0$, and at most $\pi/2$ because $B(a,1) \subset \{x \ge 0\}$. The other pieces lie in $S_\alpha \cup [e,p]$.

Self-approach. Every point z after the radial segment satisfies $z \cdot q \le 1-\varepsilon$: on the arc $z \cdot q = (1-\varepsilon)\cos(\theta-\alpha)$, and on the rectangle $z \cdot q \le t\cos(\frac{\pi}{3}-\alpha)$ because $\nu \cdot q = \cos(\frac{5\pi}{6}-\alpha) \le 0$. For $x = \lambda q$ on the segment, $\lambda \in [1-\varepsilon, 1]$, this gives $(z-x)\cdot(-q) = \lambda - z\cdot q \ge 0$, so the distance to z decreases along the segment. For the arc and the cascade, the argument of Theorem 5.5 applies verbatim: a point of the rectangle has the form $tp + y\nu$ with $t \in [\varepsilon, 1-\varepsilon]$ and $y \ge 0$, so its argument lies in $[\frac{\pi}{3}, \frac{5\pi}{6}] \subset [\frac{\pi}{3}, \alpha+\pi]$, and Lemma 5.4, scaled to radius $1-\varepsilon$, applies to the arc from $(1-\varepsilon)q$ to $(1-\varepsilon)p$. The final segment continues the cascade along its base line beyond the last cusp. For x on the last half-cycloid, with end point c^*, progress direction u, and $z = c^* + \tau u$, we have $(z-x)\cdot T(x) = (c^*-x)\cdot T(x) + \tau\, u \cdot T(x) \ge 0$, because the tangent of a half-cycloid always has a nonnegative component along u. For earlier pieces the bound $Q \ge 0$ from the proof of Lemma 5.1 applies.

(ii) The arc about a is self-approaching by Lemma 5.3. The chord from its endpoint to e lies in its future cone by Lemma 5.4, applied with centre a. The cascade lies on the side of the chord facing the arc, and the modification near the ends is as in (i). The chord has length $2\sin(\frac{\pi}{6}-\frac{\beta}{2})$, and $g'(\beta) = 1 - 2\cos(\frac{\pi}{6}-\frac{\beta}{2}) \le 1-\sqrt{3} < 0$. □

Part (ii) reflects a general phenomenon: the outer arc of K is centred at the earlier point a, and such arcs always carry slack.

7.2. **Saturated parameters.**

Lemma 7.5 (Saturated arcs have later centres)**.** *Suppose $C|_{[s_0,s_1]}$ is an arc of a circle with centre c, of angle less than π, and some $s \in (s_0, s_1)$ is saturated. Then $c \in C([s_1, L])$. In particular, an arc whose centre is not a later point of the curve has $\sigma > 0$ at all of its interior parameters; an example is the outer arc of K, whose centre is the starting point a.*

Proof. Let $\theta(\cdot)$ be the angular coordinate on the arc, increasing in time. By Lemma 5.4, after scaling, every point of $C([s_1, L])$ lies in $V(s') := c + \{\rho\,\mathbf{e}(\psi) : \theta(s_1) \le \psi \le \theta(s') + \pi\}$ for each $s' \in [s_0, s_1]$. At the saturated parameter s, one edge of $W(s)$ is a normal ray at $C(s)$. It is the ray through c, because the later points of the arc lie strictly on the side of c of the tangent line. Choose later points z_k whose directions from $C(s)$ tend to that edge. They stay away from $C(s)$, since arc points near $C(s)$ have directions near $T(s)$ and $C(s) \notin V(s)$. So a subsequence converges to a later point $z \ne C(s)$ on the normal line $\ell(s)$. The line $\ell(s)$ meets the circle only at angles $\theta(s)$ and $\theta(s)+\pi$, so z is not on the arc and $z \in C([s_1, L])$. For $s' \in [s_0, s)$ the cone $V(s')$ meets $\ell(s)$ only in c. Hence $z = c$. □

Lemma 7.6 (Contact points)**.** *Let s be saturated, and let $n(s)$ be the unit normal with $W(s) = \{\lambda T(s) + \mu n(s) : \lambda, \mu \ge 0\}$. Then the normal line $\ell(s)$ supports $H(s)$, and $H(s) \cap \ell(s)$ is a segment $[C(s), z(s)]$ with $z(s) = C(s) + d(s)n(s)$, $d(s) \ge 0$, and $z(s) \in C([s, L])$. If $z(s) \equiv c$ on an interval, the curve is a circular arc about the later point c there. In general $s \mapsto z(s)$ runs over later pieces of the curve itself.*

Proof. $H(s) \subset C(s) + W(s)$, and the line $\ell(s)$ meets this wedge exactly in the edge ray $C(s) + \mathbb{R}_+ n(s)$. Hence $\ell(s)$ supports $H(s)$, and $H(s) \cap \ell(s)$ is a segment on that

ray starting at $C(s)$. Its far endpoint is an extreme point of $H(s) = \operatorname{conv} C([s, L])$, so it belongs to $C([s, L])$. If $z(s) \equiv c$, then $(C(s) - c) \cdot T(s) = 0$ almost everywhere, so $|C(s) - c|$ is constant. □

Remark 7.7 (The cascade). In the cascade of Lemma 5.1, the normal of D_k at $c(\varphi)$ passes through its centre of curvature $\tilde{c}(\varphi)$, a later point of E_k. By Lemma 2.2 the wedge W at $c(\varphi)$ is therefore exactly the right angle spanned by the tangent and this normal, and the parameter is saturated, with contact point $z = \tilde{c}(\varphi)$. By the glide reflection the same holds on E_k, with contact points on D_{k+1}. So $z(s)$ runs over the next half-cycloid: every evolute is a piece of the curve itself. Only the last piece, which has no successor, carries slack; its length $2w/n$ tends to 0.

Lemma 7.8 (Strips)**.** *If $s_1, s_2 \le s$, $T(s_1) = \nu$ and $T(s_2) = -\nu$ for a unit vector ν, then $C([s, L]) \subset \{z : C(s_1) \cdot \nu \le z \cdot \nu \le C(s_2) \cdot \nu\}$. Consequently, if a sequence of curves contains cascades of height $h_k \to 0$ on segments $[x_k, y_k] \to [x, y]$, every uniform limit continues after y only along the ray from y in direction $y - x$.*

Proof. The first statement is the half-plane property of Lemma 2.2, used at s_1 and s_2. In a cascade the tangents at the starting points of the pieces are $\pm\nu$, perpendicular to $y - x$, on the two long sides, and at the cusps the tangent is $(y - x)/|y - x|$. The remaining curve is therefore confined to a strip of width h_k beyond the last cusp. □

A cascade of vanishing height can therefore stand only at the end of a curve, up to a collinear continuation, which adds no more length than a cascade on the extended segment would. Oscillations whose tangents stay away from the normal of their segment behave differently.

Remark 7.9 (Interior oscillations). Let $\varphi \in (0, \pi/4]$, let d be a unit vector, and let Z be a zig-zag of equal straight pieces, alternately in the directions of d rotated by $+\varphi$ and by $-\varphi$, between two parallel lines at distance h. Any two pieces make an angle 0 or $2\varphi \le \pi/2$, so all inner products of pieces are nonnegative, and the criterion (8.1) below shows that Z is self-approaching. The same computation shows that Z may be followed by any self-approaching curve that starts at the last point y of Z and stays in the wedge $y + \{w : \angle(w, d) \le \pi/2 - \varphi\}$. Up to its end pieces, Z has $1/\cos\varphi$ times the length of the segment it follows, whatever h is, and it converges uniformly to that segment as $h \to 0$. Vanishing oscillations of multiplicity $\sqrt{2}$, the value for $\varphi = \pi/4$, can thus stand in the interior of a limit curve and be followed by a turn of up to $\pi/4$. Oscillations whose later points all lie on one side reach larger multiplicities, and can also be followed by turns (Theorem 6.3).

7.3. **The cap problem.** For $\alpha \in (0, \pi/3]$ let κ_α be the supremum of the lengths of self-approaching curves in R_α from $q = \mathbf{e}(\alpha)$ to e, and let $\kappa := \kappa_{\pi/3}$, the supremal length of a self-approaching curve in the cap K from p to e.

Proposition 7.10. *For $\alpha \in (0, \pi/3]$,*

$$\sup\{\text{length}(C) : C \in \mathcal{A}_\alpha\} = \alpha + \kappa_\alpha,$$

$$\frac{\pi}{3} - \alpha + \kappa \le \kappa_\alpha \le 1 + \frac{2\pi}{3} - \alpha, \qquad 2 \le \kappa \le 1 + \frac{\pi}{3}.$$

In particular $C_{\text{diam}} \ge \frac{\pi}{3} + \kappa$.

Proof. By Lemma 7.2, $C|_{[\alpha,L]}$ is a self-approaching curve in R_α from q to e, which gives $\le$. Conversely, let Γ be such a curve and prepend the unit arc about e from a to q. Every point of R_α other than e has argument in $[\alpha,\pi]$, so by Lemma 5.4 the distance from the arc to every point of Γ is nonincreasing along the arc. Triples inside the arc or inside Γ are covered by Lemma 5.3 and by assumption; if t_1 lies on the arc and $t_2 \le t_3$ on Γ, then $|C(t_2)-C(t_3)| \le |q-C(t_3)| \le |C(t_1)-C(t_3)|$. The curve lies in $B(a,1)$ and contains e, so by Lemma 2.1 its diameter is 1 and it belongs to $\mathcal{A}_\alpha$; this gives $\ge$. The bound $\kappa \ge 2$ is Lemma 7.4(i) with $\alpha=\pi/3$. For the lower bound on κ_α, run along $\partial B(e,1)$ from q to p and then along a curve in K from p to e; the same argument, with Lemma 5.4 applied to the arc from q to p, whose future cone $[\frac{\pi}{3},\alpha+\pi]$ contains the arguments $[\frac{\pi}{3},\frac{\pi}{2}]$ of K, shows that this is a self-approaching curve in R_α. The upper bounds follow from Corollary 3.2, as in Proposition 7.3, with $\operatorname{per} R_\alpha = 1+\frac{2\pi}{3}-\alpha$ and $\operatorname{per} K = 1+\frac{\pi}{3}$. Finally $C_{\mathrm{diam}} \ge \frac{\pi}{3}+\kappa_{\pi/3}$ by the first statement with $\alpha=\pi/3$. □

The straight cascade gives $\kappa \ge 2$, and the arc-and-cascade construction corresponds to it. The hull inequality gives $\kappa \le \operatorname{per} K$, and this is the true value.

Theorem 7.11. $\kappa = 1+\dfrac{\pi}{3}$.

Proof. By Proposition 7.10 it remains to show $\kappa \ge 1+\pi/3$. The outer arc of K is $\{q(\theta) : \frac{\pi}{3} \le \theta \le \frac{\pi}{2}\}$ with $q(\theta) = 2\cos\theta\,\mathbf{e}(\theta)$, and $q(\theta)-a = \mathbf{e}(2\theta)$; so $q(\pi/3)=p$, $q(\pi/2)=e$, and $q(\theta)$ lies on $\partial B(a,1)$ at the angle 2θ about a. Fix $m \ge 1$ and put $\theta_i = \frac{\pi}{3}+\frac{i\pi}{6(m+1)}$ and $q_i = q(\theta_i)$ for $0 \le i \le m+1$. Let $\ell_i = |q_{i+1}-q_i|$, $d_i = (q_{i+1}-q_i)/\ell_i$, and let n_i be the unit normal of d_i on the side of a. Consecutive chords turn by 2Δ with $\Delta = \frac{\pi}{6(m+1)}$, and $\Pi = \operatorname{conv}\{q_0,\dots,q_{m+1}\} \subset K$.

The angle Θ_i. For x on the chord $[q_i,q_{i+1}]$ let $\Theta_i(x)$ be the angle between d_i and $e-x$, measured towards n_i. Seen from x, which lies inside the circle $\partial B(a,1)$, the points $q_{i+1},q_{i+2},\dots,q_{m+1}=e$ appear at angles from d_i that increase from 0 to $\Theta_i(x)$, so $\Theta_i(x)$ is the angle under which $\Pi_{i+1} = \operatorname{conv}\{q_{i+1},\dots,q_{m+1}\}$ is seen from x. As x moves along d_i, $\Theta_i(x)$ increases, up to the angle $\frac{\pi}{2}-\theta_i \le \frac{\pi}{6}$ between the chord and $[q_{i+1},e]$ at q_{i+1}, and $\frac{d}{ds}|x-e| = -\cos\Theta_i(x)$. Hence

$$\begin{aligned} \int_{[q_i,q_{i+1}]} \bigl(1+\cos\Theta_i\bigr)\,ds &= \ell_i + |q_i-e| - |q_{i+1}-e|, \\ \sum_{i=0}^{m} \int_{[q_i,q_{i+1}]} \bigl(1+\cos\Theta_i\bigr)\,ds &= 1+\sum_{i=0}^{m}\ell_i. \end{aligned} \tag{7.1}$$

The left-hand side is the length of an oscillation of multiplicity $1+\cos\Theta_i$ along each chord, and the right-hand side is $\operatorname{per}\Pi$.

The curve. Let $R>0$ be small. For $0 \le i \le m$ the curve Γ runs straight from q_i to q_i+4Rd_i; then along the image of a cut cascade $Z(R;\Theta_{i,0},\Theta_{i,1},\dots)$ under $z \mapsto q_i+4Rd_i+z_1d_i+z_2n_i$, with as many periods as fit before q_{i+1}, and before $q_{m+1}-4Rd_m$ on the last chord; and then straight to q_{i+1}. Let $y_{i,k}$ be the end of the k-th period on the i-th chord. For $i<m$ let E_i be the union, over $j>i$, of the segments $[q_j,q_j+4Rd_j]$ and the rectangles $\{q_j+\sigma d_j+\rho n_j : 4R \le \sigma \le \ell_j,\ 0 \le \rho \le R\}$, and let $\Theta_{i,k}$ be the largest angle between d_i and $z-y_{i,k}$, $z \in E_i$, measured towards n_i. On the last chord put $\Theta_{m,k}=0$. By Lemma 6.2(ii) every point of Γ after the i-th chord lies in E_i.

The angles are admissible. All of E_i lies on the side of n_i of the line through the i-th chord, so $\Theta_{i,k} \geq 0$. If $|y_{i,k} - q_{i+1}| \geq \delta$, then $E_i \subset \Pi_{i+1} + B(0,R)$ and $\operatorname{dist}(y_{i,k}, \Pi_{i+1}) = |y_{i,k} - q_{i+1}| \geq \delta$, so

$$\Theta_{i,k} \leq \Theta_i(y_{i,k}) + \arcsin(R/\delta). \tag{7.2}$$

Near q_{i+1} the largest angle comes from the start of the next rectangle; with $\sigma \geq 4R \geq 4\rho$ and a turn of $2\Delta \leq \pi/6$ it is at most $\arctan \frac{4\sin(\pi/6)+1}{4\cos(\pi/6)-1} < \frac{\pi}{2}$, and the rectangles of the chords after the next are seen under angles at most $\frac{\pi}{6} + O(R)$. So $\Theta_{i,k} < \pi/2$ for small R.

$\Gamma \subset K$. By Lemma 6.2(ii) the cascade on the i-th chord lies in $\{q_i + \sigma d_i + \rho n_i : 4R \leq \sigma \leq \ell_i,\ 0 \leq \rho \leq \frac{\pi}{4}R\}$. The interior angles of Π are $\frac{\pi}{6} - \Delta \geq \frac{\pi}{12}$ at p and at e, where $4\tan\frac{\pi}{12} > \frac{\pi}{4}$, and larger than $\pi/2$ at $q_1, \ldots, q_m$. Hence these rectangles lie in $\Pi \subset K$ for small R.

Γ *is self-approaching.* We check the half-plane criterion at a point x of the i-th chord. If x lies on a straight piece, the later points z of the same chord satisfy $(z-x)\cdot d_i \geq 0$ by Lemma 6.2(ii). A later point on the j-th chord, $j > i$, has the form $z = q_j + \sigma d_j + \rho n_j$ with $\sigma \geq 0$, $0 \leq \rho \leq \frac{\pi}{4}R$, and $\rho = 0$ if $\sigma < 4R$. The angle between d_i and $q_j - q_{i+1}$ is at most $\pi/6$, and d_j and n_j are d_i and n_i turned by at most $\pi/3$, so $(z-x)\cdot d_i \geq (q_j - q_{i+1})\cdot d_i + \frac{\sigma}{2} - \rho\frac{\sqrt{3}}{2} \geq 0$. If x lies on the k-th period of the cascade, later points of the same cascade are covered by Lemma 6.2(iii). Every other later point lies in $y_{i,k} + V_{\Theta_{i,k}}$ in the frame (d_i, n_i): points of the final straight piece of the chord lie on the ray from $y_{i,k}$ in direction d_i, and points of later chords lie in E_i. Lemma 6.2(iv) covers them. On the last chord the only later points outside its cascade lie on the final segment to e, in direction d_m, and $\Theta_{m,k} = 0$.

The length. Each period on the i-th chord has length $R(1+\cos\Theta_{i,k})$ and advances by R, so $\operatorname{length}(\Gamma) = \sum_i \ell_i + \sum_{i,k} R\cos\Theta_{i,k}$. Fix $\delta > 0$. By (7.2), and since $\cos\Theta_i$ decreases along the chord and $y_{i,k}$ is the end of a period,

$$\sum_k R\cos\Theta_{i,k} \geq \int_{[q_i, q_{i+1}]} \cos\Theta_i \, ds - \delta - 10R - \ell_i \arcsin(R/\delta).$$

Letting $R \to 0$ and then $\delta \to 0$, (7.1) gives $\liminf_{R\to 0} \operatorname{length}(\Gamma) \geq 1 + \sum_i \ell_i$. Each Γ is a self-approaching curve in K from p to e, so $\kappa \geq 1 + \sum_i \ell_i = 1 + 2(m+1)\sin\Delta$, and $m \to \infty$ gives $\kappa \geq 1 + \pi/3$. □

Corollary 7.12. *For every* $\alpha \in (0, \pi/3]$, $\kappa_\alpha = 1 + \frac{2\pi}{3} - \alpha$ *and*

$$\sup\{\operatorname{length}(C) : C \in \mathcal{A}_\alpha\} = 1 + \frac{2\pi}{3}.$$

In particular $C_{\mathrm{diam}} \geq 1 + \frac{2\pi}{3}$.

Proof. Proposition 7.10 gives $\frac{\pi}{3} - \alpha + \kappa \leq \kappa_\alpha \leq 1 + \frac{2\pi}{3} - \alpha$, and by Theorem 7.11 both sides are equal. The supremum is $\alpha + \kappa_\alpha$, and the curves of $\mathcal{A}_\alpha$ have diameter one. □

Remark 7.13 (The limit curve). As $R \to 0$ and then $m \to \infty$, the curves of Theorem 7.11 converge uniformly to the outer arc of K, traversed with the multiplicity $1 + \cos(\beta/2)$, where β is the angle about a of the part of the arc that remains; indeed $\int_0^{\pi/3}(1 + \cos\frac{\beta}{2})\,d\beta = \frac{\pi}{3} + 1$. They do not contradict Lemma 7.5, which concerns curves that contain an arc about an earlier point: length is only lower semicontinuous (Remark 3.4). Nor do they contradict Lemma 7.8: at each vertex

q_{i+1} the oscillation turns by 2Δ, which is smaller than its cut angles, and cut cascades admit such turns by Lemma 6.2(iv). Their argument about e decreases on every arc U, so Proposition 7.14 does not apply to them.

7.4. **Curves that wind monotonically about their endpoint.** The curves of Section 5 have the following form: a curve that winds monotonically about the endpoint f, followed by a cascade of vanishing height on a final segment $[y, f]$, whose length in the limit is $2|y - f|$ (Lemma 5.1). No curve of this form beats $2 + \pi/3$.

Proposition 7.14 (Monotone winding)**.** *Let C be a self-approaching curve with $\operatorname{diam} C = 1$, starting point a and endpoint f. Suppose that C consists of a curve C_1 from a to a point y, followed by the segment $[y, f]$, and that the argument of C_1 about f is monotone and varies by at most π. Then*

$$\operatorname{length}(C_1) + 2|y - f| \le 2 + \frac{\pi}{3}.$$

Proof. If C passes through f, it stays there by (1.1), so C_1 meets f at most at its end. Take polar coordinates about f with $a = f + \rho_0 \mathbf{e}(0)$, where $\rho_0 = |a - f| \le 1$, and write $C_1(s) = f + \rho(s)\,\mathbf{e}(\theta(s))$ with a continuous argument θ, nondecreasing after a reflection, from 0 to some $\theta_1 \le \pi$. Since f is a later point of C_1, (1.1) makes ρ nonincreasing, from ρ_0 to $\rho_1 := |y - f|$. From $|C_1'|^2 = \rho'^2 + \rho^2\theta'^2 \le (|\rho'| + \rho\,\theta')^2$ we get

$$\operatorname{length}(C_1) \le \rho_0 - \rho_1 + \int \rho\, d\theta.$$

By Lemma 2.1, $C_1 \subset B(a, 1) \cap B(f, \rho_0)$. The points $f + r\,\mathbf{e}(\theta)$, $r \ge 0$, of this set are those with $r \le R(\theta) := \min\bigl(\rho_0,\ \rho_0 \cos\theta + \sqrt{1 - \rho_0^2 \sin^2\theta}\bigr)$. Since θ is nondecreasing, $\int \rho\, d\theta \le \int_0^{\theta_1} R\, d\theta$, and $\rho_1 \le R(\theta_1)$. Hence

$$\operatorname{length}(C_1) + 2\rho_1 \le F(\theta_1) := \rho_0 + R(\theta_1) + \int_0^{\theta_1} R(\theta)\, d\theta.$$

If $\rho_0 \le \frac{1}{2}$, then $B(f, \rho_0) \subset B(a, 1)$, so $R \equiv \rho_0$ and $F(\theta_1) \le \rho_0(2 + \pi) < 2 + \frac{\pi}{3}$. Let $\rho_0 \in (\frac{1}{2}, 1]$, and let $\theta_* = 2 \arcsin \frac{1}{2\rho_0} \in [\frac{\pi}{3}, \pi)$ be the argument at which $\partial B(f, \rho_0)$ meets $\partial B(a, 1)$. Then $R \equiv \rho_0$ on $[0, \theta_*]$, so for $\theta_1 \le \theta_*$

$$F(\theta_1) \le \rho_0(2 + \theta_*) = \frac{2 + \theta_*}{2 \sin(\theta_*/2)} = \Phi(\theta_*),$$

with Φ as in Proposition 5.6. By the proof of that proposition, the logarithmic derivative of Φ on $[\frac{\pi}{3}, \pi)$ has the sign of $\tan u - u - 1$ with $u = \theta_*/2$, an increasing function; so $\Phi(\theta_*) \le \max\bigl(\Phi(\frac{\pi}{3}), \frac{2+\pi}{2}\bigr) = 2 + \frac{\pi}{3}$.

For $\theta_1 \in [\theta_*, \pi]$ we follow $\partial B(a, 1)$. Its points $P(\psi) = a + \mathbf{e}(\psi)$ with $\psi \in [\psi_*, \pi]$, where $\cos\psi_* = -1/(2\rho_0)$, have arguments about f increasing from θ_* to π (to $\pi/2$ if $\rho_0 = 1$, where R vanishes beyond), and moduli $g(\psi) = (1 + \rho_0^2 + 2\rho_0 \cos\psi)^{1/2}$. Along this arc $R\, d\theta = (1 + \rho_0 \cos\psi)\, g^{-1} d\psi$, so as a function of the parameter ψ_1 of the point at argument θ_1,

$$\frac{dF}{d\psi_1} = g'(\psi_1) + \frac{1 + \rho_0 \cos\psi_1}{g(\psi_1)} = \frac{1 + \sqrt{2}\,\rho_0 \cos(\psi_1 + \pi/4)}{g(\psi_1)}.$$

Since $\psi_* \ge 2\pi/3$, the numerator first decreases and then increases on $[\psi_*, \pi]$. Its value at ψ_* is $\frac{1}{2} - (\rho_0^2 - \frac{1}{4})^{1/2}$, and its minimum is at least $1 - \sqrt{2}\,\rho_0$. If $\rho_0 < 1/\sqrt{2}$ it is therefore positive throughout, and otherwise it is nonpositive and then nonnegative.

In both cases F attains its maximum over $[\theta_*, \pi]$ at θ_*, where it is $\Phi(\theta_*)$, or at π. Finally, $\sqrt{1-t} \le 1 - t/2$ gives $R(\theta) \le 1 + \rho_0 \cos\theta - \frac{\rho_0^2}{2}\sin^2\theta$ on $[\theta_*, \pi]$, and integrating, with $\rho_0 = 1/(2\sin x)$ and $\theta_* = 2x$, $x \in [\frac{\pi}{6}, \frac{\pi}{2})$,

$$F(\pi) = 1 + \rho_0\theta_* + \int_{\theta_*}^{\pi} R\,d\theta \le 1 + \frac{x}{\sin x} + (\pi - 2x) - \cos x - \frac{\pi - 2x}{16\sin^2 x} - \frac{\sin 4x}{32\sin^2 x}.$$

Since $\sin^2 x \ge \frac{1}{4}$, the right-hand side is at most $\frac{9}{8} + \frac{x}{\sin x} + (\pi - 2x)\big(1 - \frac{1}{16\sin^2 x}\big) - \cos x$. Here $x/\sin x$, $1 - \frac{1}{16\sin^2 x} \ge 0$ and $-\cos x$ are nondecreasing, and $\pi - 2x \ge 0$ is decreasing. Bounding each of them on each of 24 equal subintervals of $[\frac{\pi}{6}, \frac{\pi}{2}]$ by its value at the appropriate endpoint gives $F(\pi) \le 3.0029 < 2 + \frac{\pi}{3}$; twelve subintervals give only 3.0745. □

Corollary 7.15. *Let $\alpha \in (0, \pi/3]$, and let Γ be a self-approaching curve in R_α from q to e that consists of a curve Γ_1 from q to a point y, followed by the segment $[y, e]$. If the argument of Γ_1 about e is nondecreasing, then*

$$\operatorname{length}(\Gamma_1) + 2|y| \le 2 + \frac{\pi}{3} - \alpha.$$

In particular, for $\alpha = \pi/3$: every self-approaching curve in K from p to e whose argument about e is nondecreasing before its last segment $[y, e]$ satisfies $\operatorname{length}(\Gamma_1) + 2|y| \le 2$.

Proof. Prepend the arc of $\partial B(e, 1)$ from a to q. By Lemma 5.4 with $f = e$, the distance from this arc to every point of R_α is nonincreasing along the arc, so the resulting curve is self-approaching; it lies in $B(a, 1)$, so its diameter is 1; and its argument about e is nondecreasing from 0 to at most $\pi/2$. Proposition 7.14 gives $\alpha + \operatorname{length}(\Gamma_1) + 2|y| \le 2 + \pi/3$. □

Remark 7.16 (The hypotheses). The proofs of Proposition 7.14 and Corollary 7.15 use self-approach only through the distance to the endpoint, which is nonincreasing; every other constraint enters through the region. Both hypotheses are essential to this method. If the argument may vary by more than π, the relaxation F exceeds $2 + \pi/3$ once the variation exceeds about 217°. If the argument may decrease, the bound itself fails: the argument about e of the curves of Theorem 7.11 decreases on every arc U of their cut cascades, and their length in K tends to $1 + \pi/3 > 2$. Lemma 7.5 shows that the outer arc of K cannot be followed without slack, and Lemma 7.8 shows that a cascade of vanishing height cannot be followed by a turn. These curves avoid both: they follow the outer arc only in the limit, through cut cascades along inscribed chords, and cut cascades can be followed by turns (Remark 7.13).

8. Polygons: a certificate, and the search that preceded it

8.1. **A polygonal model.**

Lemma 8.1 (Polygons)**.** *A polygon $v_0, \dots, v_N$ is self-approaching if and only if*

$$(v_j - v_i) \cdot (v_i - v_{i-1}) \ge 0 \qquad \text{for all } 1 \le i < j \le N. \tag{8.1}$$

Proof. Along a segment, the squared distance to a fixed point is a convex quadratic in the parameter, so it is nonincreasing on the segment if and only if its derivative at the end of the segment is ≤ 0. Later points of the same segment are approached

trivially. Every later point is a convex combination of two consecutive later vertices, and (8.1) defines a half-plane. □

A cascade of vanishing height doubles the length of a segment (Lemma 5.1), and by Lemma 7.8 it can only stand in terminal position. We therefore maximize $\Lambda(v)$ of (5.1) subject to (8.1) and a region constraint. By Lemma 5.2, a feasible v whose inequalities (8.1) hold strictly and whose last segment lies in the interior of the region, apart from its endpoint, gives the lower bound $\Lambda(v)$ for the supremum. Both conditions are needed: $v = (-1,0),(0,0),(\frac{1}{2},-\frac{1}{2}),(\frac{1}{2},-1),(0,-1),(0,-2)$ satisfies (8.1), but its first segment forces later points into $\{x \geq 0\}$ and its fourth into $\{x \leq 0\}$, so its last segment cannot carry a cascade. Interior oscillations, such as those of Remark 7.9, enter the model only through polygon vertices. We consider three problems.

(P1) *Cap.* $v_0 = p$, $v_N = e$, all $v_i \in K$. Its supremum is $\kappa = 1 + \pi/3$ (Theorem 7.11); the straight cascade gives 2.

(P2) *Early exit.* $v_0 = q = \mathbf{e}(\alpha)$, $v_N = e$, all $v_i \in R_\alpha$, for $\alpha \in \{0°, 15°, 30°, 45°\}$. Its supremum is κ_α; Lemma 7.4(i) gives $2 + \pi/3 - \alpha$. For $\alpha = 0$ the region is the upper half of the lens.

(P3) *Global.* $v_0 = a = 0$ and $|v_i| \leq 1$, which fixes the diameter at 1 by Lemma 2.1. The family $C_{\pi/3,n}$ gives $2 + \pi/3$.

8.2. **The certificate.** Theorem 7.11 determines κ, so polygons are not needed for the lower bound. We keep one as a finite check that does not depend on Section 6; it can be verified by a short program.

Theorem 8.2. *There is a polygon $v_0 = p, v_1, \ldots, v_{362} = e$ in K with rational vertices that satisfies (8.1) with strict inequalities, whose last segment lies in the interior of K apart from its endpoint e, and whose value (5.1) is*

$$\Lambda_\# = 2.0096799432101572165\ldots$$

Consequently $\kappa \geq \Lambda_\# > 2$, by a computer-assisted proof that does not use cut cascades.

Proof. Rotate the frame of Section 7 so that the chord ep is horizontal: $e = (0,0)$, $p = (1,0)$ and $a = (\frac{1}{2}, -\frac{\sqrt{3}}{2})$. Then $K = \{(x,y) : y \geq 0,\ \mu \geq \sqrt{3}\,y\}$ with $\mu = x - x^2 - y^2$, so a point with rational coordinates lies in K if and only if $y \geq 0$, $\mu \geq 0$ and $\mu^2 \geq 3y^2$. The vertices of the polygon are rational. The containment conditions for all 363 vertices and the $\binom{362}{2} = 65341$ inequalities (8.1), each strict, were verified in exact rational arithmetic, and Λ was evaluated to 50 digits. The vertex v_{361} satisfies $y > 0$ and $\mu^2 > 3y^2$, so it is an interior point of K; since $e \in \partial K$ and K is convex, the open segment (v_{361}, e) lies in the interior of K. Strict inequalities for finitely many pairs give (5.2) with some $\eta > 0$; the computed smallest value of $\cos\angle(v_j - v_i, v_i - v_{i-1})$ is $1.96 \cdot 10^{-6}$. Lemma 5.2 gives self-approaching curves in K from p to e of length at least $\Lambda_\# - 2\varepsilon$ for every $\varepsilon > 0$. □

The polygon, and a program that performs these checks in rational arithmetic, accompany the paper. Figure 5 shows it. It starts along the chord at p, oscillates along the chord for $x \in [0.901, 1]$ at heights below 0.013, climbs to $v_{361} \approx (0.89997, 0.05050)$ on the outer arc, and ends with the radius from v_{361} to e, of length 0.9014, which carries the cascade. Its length before the last segment is 0.2069. Its oscillations have the shape of cut cascades (Figure 3), with steep rises, chords, and descents, and

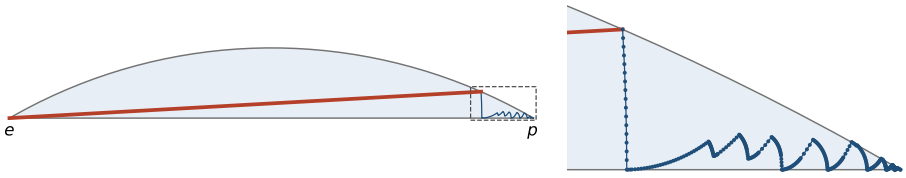


FIGURE 5. The polygon of Theorem 8.2 in the cap K (shaded) with the chord ep horizontal. Left: the whole polygon; the thick last segment carries the cascade. Right: the dashed box enlarged, with the vertices marked. The polygon oscillates along the chord near p before it climbs to the outer arc.

Θ	30°	37.5°	45°	52.5°	60°	67.5°	75°	82.5°
polygon, $P = 12$	1.7564	1.7102	1.6402	1.5666	1.4703	1.3628	1.2508	1.1274
polygon, $P = 24$	1.8081		1.6745		1.4858			
cut cascade	1.8660	1.7934	1.7071	1.6088	1.5000	1.3827	1.2588	1.1305
best zig-zag	1.4142	1.4142	1.4142	1.4021	1.3660	1.3066	1.2247	1.1220

TABLE 1. Multiplicity of periodic self-approaching oscillations with unit tangents in $\mathcal{T}_\Theta$: polygons with P vertices per period, validated over 12 periods to 10^{-12} (for $P = 24$ started from the cut cascade); the exact value $1 + \cos\Theta$ of Theorem 6.3, attained by the cut cascade; and the best symmetric zig-zag.

its argument about e decreases 176 times. It was obtained from a polygon with 92 segments and value 2.0030889, found by local optimization started from one-sided oscillations along the chord, by inserting the midpoints of all segments but the last twice, which preserves (8.1), and re-optimizing after each step by sequential linear programming with the normalized margin 10^{-6}; then the vertices were rounded to rationals.

Remark 8.3 (Polygons and cut cascades). Polygons approach the cut cascades slowly. Table 1 lists the largest multiplicity of periodic self-approaching polygons with P vertices per period whose segment directions lie in $\mathcal{T}_\Theta$, the polygonal version of Theorem 6.3. At $\Theta = 30°$ they reach 1.7564, 1.7792 and 1.8081 for $P = 12$, 16 and 24, so P times the gap to $1 + \cos\Theta = 1.8660$ stays near 1.4. A polygon must create slack at every saturated pair of the cascade, where a curve needs none, and this costs a fixed fraction of the multiplicity per vertex. The same effect makes the polygon of Theorem 8.2 fall short of $\kappa = 1 + \pi/3$.

8.3. **The search that preceded it.** For each problem and each N we ran local optimizations (sequential quadratic programming with analytic Jacobians, refined by an augmented Lagrangian method) from many random and structured starting configurations. A solution was accepted only if, after merging vertices closer than 10^{-6}, every constraint (8.1) held in the scale-invariant form $\cos\angle(v_j - v_i, v_i - v_{i-1}) \geq$

problem	N	runs	acc.	search	reference	limit
cap K	12	30	23	2.000000	2.000000	2.0000
cap K	20	30	16	2.000000	2.000000	2.0000
cap K	30	20	10	2.000000	2.000000	2.0000
cap K, Thm. 8.2	362			2.009680		2.0000
R_{0°	20	20	18	2.990253	2.990253	3.0472
R_{15°	20	20	16	2.755565	2.755565	2.7854
R_{30°	20	20	14	2.510639	2.511329	2.5236
R_{45°	20	20	13	2.254069	2.258982	2.2618

TABLE 2. Best accepted values of Λ in the cap and in the regions R_α. The reference is the exact optimum of the polygonal version of Lemma 7.4: a straight cascade in the cap, and in R_α a spiral polygon about e from q followed by the terminal cascade on the radius. The last column is the limit $2+\pi/3-\alpha$ of Lemma 7.4(i), which is 2 for the cap. The local search did not exceed it; the polygon of Theorem 8.2 does. The true suprema are $1+2\pi/3-\alpha$ (Corollary 7.12), 2.0472 for the cap.

N	runs	acc.	search	construction	independent
12	33	33	2.947368	2.947368	2.947368
20	33	30	2.990253	2.990253	2.990253
30	30	24	3.010147	3.010147	3.007364
40	23	16	3.019739	3.019739	2.924206

TABLE 3. The global problem (P3). The search optimum is compared with the optimum of the polygonal version of $C_{\pi/3,n}$, computed independently. The last column is the best value reached from random point clouds and random walks alone.

-10^{-6} and the region constraints held to 10^{-6}. The scale-invariant form is essential: microscopic zig-zags can violate (8.1) by only 10^{-7} while violating it by order one after normalization, and such configurations raised the cap value to 2.0084 before they were excluded. Tables 2 and 3 list the best accepted values. None of the starting configurations contained oscillations.

In the cap, 49 of 80 runs were accepted, and none exceeded 2 by more than 10^{-9}. In 42 of them the polygon reduced to the single segment from p to e with its cascade, and the largest value among the others was 1.999999. Started from one-sided oscillations along the chord, a local optimization of the same kind exceeds 2; the polygon of Theorem 8.2 was obtained in this way and then refined. In R_α the search never exceeded the reference and differed from it by at most $5\cdot10^{-3}$; it found the spiral and the terminal cascade from starting configurations that did not contain them. In the global problem the search optimum agrees to within $5\cdot10^{-10}$ with the optimum of the polygonal version of $C_{\pi/3,n}$, a spiral polygon about the endpoint followed by the terminal cascade (Figure 6). That optimum increases to $2+\pi/3$: its gap is 0.00534 at $N=200$, and N times the gap decreases from 1.198 at $N=12$ to 1.068 at $N=200$. The reference configurations of (P1) and (P2) wind monotonically about e before their last segment, and so, as far as Figure 6 shows,

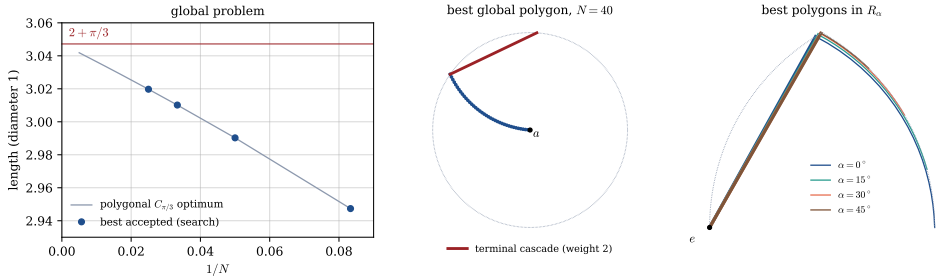


FIGURE 6. Left: best accepted values of the global problem against $1/N$ (dots), and the optimum of the polygonal construction (line), which tends to $2+\pi/3$. Middle: the best global polygon for $N=40$, a spiral polygon about the endpoint followed by the terminal cascade (thick). Right: the best polygons in R_α, each ending with its terminal cascade.

n	length	diameter	ratio	max. violation
2	3.047198	1.651424	1.845194	1.1e-16
4	3.047198	1.283975	2.373253	1.1e-16
8	3.047198	1.128322	2.700646	1.1e-16
16	3.047198	1.060456	2.873479	1.1e-16
32	3.047198	1.029285	2.960499	1.1e-16
64	3.047198	1.014406	3.003922	0.0e+00

TABLE 4. The family $C_{\pi/3,n}$, built from the explicit formulas. Diameters are maxima over the samples, hence lower bounds; the last column is the largest violation of (1.1) on $4\cdot10^5$ random triples.

curve	length	per(conv C)	$\int(\cos\alpha+\cos\beta)\,ds$
$C_{\pi/3,8}$	3.047198	3.279960	3.279960
$C_{\pi/4,4}$	2.785398	3.250922	3.250922

TABLE 5. Both sides of (3.1): the perimeter of the hull of a dense sample, and Gauss–Legendre quadrature of the right-hand side with exact tangents. They agree to $6\cdot10^{-9}$.

do the best polygons found by the search. For such curves Corollary 7.15 shows that the limits in Table 2 cannot be exceeded.

8.4. **Checks.** Table 4 lists $C_{\pi/3,n}$ built from the explicit formulas of Section 5, with 200 points per half-cycloid and 2000 on the arc; condition (1.1) was tested on $4\cdot10^5$ random ordered triples. Table 5 compares the two sides of (3.1) on dense samples.

Two observations bound what such computations can show. First, the condition must hold along the curve, not only at sample points: sequences that satisfy (1.1) at their own points can be much longer than any curve, and in the cap such a sequence of 31 points has polygonal length 2.0907, which exceeds even $\operatorname{per} K = 2.0472$. Second, the search is local, and polygons approximate the relevant curves slowly. Without the terminal cascade the best polygons in the cap reach only 1.42 for

$N = 16$ and 1.55 for $N = 32$; a polygon approximating an arc about a later point must spiral inwards, at a cost of about θ^2/N; and interior oscillations appear only through vertices. A curve beating the construction by less than about $1.2/N$ goes undetected at vertex count N unless the search is started near its shape. This is how the search of Table 2 missed the polygon of Theorem 8.2, which beats 2 by 0.0097 at $N = 362$ through oscillations, and the cut cascades of Theorem 7.11, which beat 2 by 0.047 in the limit.

9. Further bounds and open problems

We have shown $1 + \frac{2\pi}{3} \le C_{\mathrm{diam}} \le \frac{2\pi}{3} + 4\sin\frac{\pi}{12}$, that is, $3.0943\ldots \le C_{\mathrm{diam}} \le 3.1296\ldots$. The lower bound is the exact supremum over curves that begin on the circle about their last diametral partner, and their hulls tend to the half-lens. The upper bound is the exact optimum of the hull problem of Theorem 4.1, and the trimmed Reuleaux triangle attains it. The two bounds differ by $\sec\frac{\pi}{12} - 1 = 0.0352\ldots$. Two observations narrow the question.

Proposition 9.1 (Curves on one side)**.** *Let C be a self-approaching curve with $\operatorname{diam} C = 1$, starting point a and last diametral partner e. If C lies in a closed half-plane bounded by the line through a and e, then $\operatorname{length}(C) \le 1 + \frac{2\pi}{3}$. The curves of the classes $\mathcal{A}_\alpha$ are of this kind, so by Corollary 7.12 this bound is the supremum of length over all such curves.*

Proof. By Lemma 2.4, C lies in the part of the lens $B(a,1) \cap B(e,1)$ on one side of the line. This half-lens is convex, and it is bounded by the segment $[a,e]$ and by two arcs of angle $\pi/3$, about e and about a; so its perimeter is $1 + \frac{2\pi}{3}$. Corollary 3.2 and the monotonicity of perimeter give the bound. In the frame of Section 7, a curve of $\mathcal{A}_\alpha$ consists of an arc with arguments in $[0,\alpha]$ and a curve in R_α, and both lie in the upper half-plane. □

A curve longer than $1 + \frac{2\pi}{3}$ times its diameter, if one exists, therefore has points strictly on both sides of the line through its starting point and its last diametral partner.

Proposition 9.2 (Pair condition)**.** *Let C be a self-approaching curve with starting point a. Any two points z, w of C satisfy*

$$|z - w| \le \max\bigl(|z - a|,\, |w - a|\bigr). \tag{9.1}$$

Let $\mathcal{K}$ be the class of compact convex sets K with $a \in K \subset B(a,1)$ whose extreme points satisfy (9.1) in pairs, and let $P_* = \sup\{\operatorname{per} K : K \in \mathcal{K}\}$. Then $C_{\mathrm{diam}} \le P_*$.

Proof. If $z = C(s)$ and $w = C(t)$ with $s \le t$, the triple $(0, s, t)$ in (1.1) gives $|z - w| \le |a - w|$. If $\operatorname{diam} C = 1$, then $\operatorname{conv} C \subset B(a,1)$ by Lemma 2.1, and the extreme points of $\operatorname{conv} C$ are points of C. So $\operatorname{conv} C \in \mathcal{K}$, and $\operatorname{length}(C) \le \operatorname{per}(\operatorname{conv} C) \le P_*$ by (1.2). □

Condition (9.1) does not refer to the order of the points. In the triangle a, z, w it says that the side opposite to a is not the only longest side, so the angle at a is at most $\pi/2$. Hence every $K \in \mathcal{K}$ has diameter at most one and lies in a wedge of opening at most $\pi/2$ at a, as in the proof of Lemma 2.2, and $P_* \le \frac{2\pi}{3} + 4\sin\frac{\pi}{12}$ by Theorem 4.1. The half-lens belongs to $\mathcal{K}$: two of its extreme points on the arc about e, at the angles $\theta_1 < \theta_2$ from a, are at distance $2\sin\frac{\theta_2 - \theta_1}{2} \le 2\sin\frac{\theta_2}{2}$, the distance

of the second from a, and every other pair contains a or a point at distance one from a. So $P_* \geq 1 + \frac{2\pi}{3}$.

Corollary 9.3. $P_* < \frac{2\pi}{3} + 4 \sin \frac{\pi}{12}$. *In particular the second inequality of Theorem 1.1 is strict.*

Proof. Fix $a = 0$, and let $K_n \in \mathcal{K}$ with $\operatorname{per} K_n \to P_*$. By the Blaschke selection theorem a subsequence converges in the Hausdorff metric to a compact convex set K with $0 \in K \subset B(0,1)$, and $\operatorname{per} K = P_*$, because perimeter is continuous [Sch14]. We show that $K \in \mathcal{K}$. Let z be an exposed point of K, so that z is the only point of K with $z \cdot u = h_K(u)$ for some direction u, and choose extreme points z_n of K_n with $z_n \cdot u = h_{K_n}(u)$. Every limit point of (z_n) lies in K and on the supporting line of K with normal u, so $z_n \to z$. Hence (9.1) holds for every pair of exposed points of K, and then for every pair of extreme points, because the exposed points are dense in the extreme points [Sch14, Section 1.4].

Suppose that $P_* = \frac{2\pi}{3} + 4 \sin \frac{\pi}{12}$. Then K attains equality in Theorem 4.1, so it is a trimmed Reuleaux triangle with its right-angled corner at a. Its vertices z, w at the far ends of the two segments through a satisfy $|z - a| = |w - a| = 2 \sin \frac{\pi}{12}$ and $|z - w| = 2\sqrt{2} \sin \frac{\pi}{12}$, which contradicts (9.1). □

So the trimmed Reuleaux triangle is not the hull of a self-approaching curve that starts at its corner, and the hull method is not exhausted by Theorem 4.1. We do not know P_*. In about 800 local optimizations over polygons with 12 and with 20 vertices that satisfy (9.1), started from random and from structured configurations, no perimeter exceeded $1 + \frac{2\pi}{3} = 3.094395$; the largest values, 3.091738 and 3.093438, belong to polygons inscribed in a half-lens.

(1) Determine C_{diam}. Is it $1 + \frac{2\pi}{3}$? By Proposition 9.1 a longer curve must have points on both sides of the line through its starting point and its last diametral partner. By Proposition 9.2 it suffices to show that $P_* = 1 + \frac{2\pi}{3}$, which is a question about convex sets alone.

(2) Give an explicit upper bound for P_* below that of Theorem 4.1, or improve the upper bound by an estimate for the defect. The cut cascades show that arcs about earlier points can be followed without slack in the limit, so an estimate for the defect must use the order in which the extreme points are visited.

(3) Determine the largest multiplicity of a vanishing self-approaching oscillation whose later points lie in a wedge whose edges make angles θ_1 and θ_2 with its direction, on both sides. The argument of Theorem 6.3 gives $\cos \theta_1 + \cos \theta_2$, which cut cascades attain for $\theta_1 = 0$ and the zig-zag for $\theta_1 = \theta_2 = \pi/4$.

Code and data. The Python scripts that produce every number and figure accompany this paper: the exact check of the dual certificate of Theorem 4.1 to 50 digits together with a numerical test of (4.1) on random polygons; the cut cascades, with a check of the half-plane criterion for random cut angles and for the curves of Theorem 7.11; the periodic polygons of Table 1; the sequential linear programming used for Theorem 8.2; and the polygonal model, the explicit cascade, and the check of the hull identity of Section 8. The search of Section 9 over polygons with the pair condition (9.1) is included. The certified polygon of Theorem 8.2 (rational vertices) and a program that checks it in exact arithmetic are included.

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